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(54) **APPARATUS FOR AEROSOL GENERATING  
DEVICE HAVING HEATING CONTROL  
CIRCUIT HEATING BASED ON  
TEMPERATURE OF THE HEATER OR  
HEATERS**

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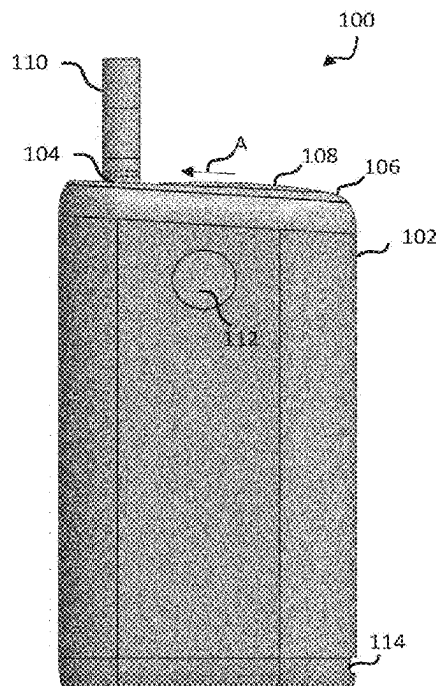
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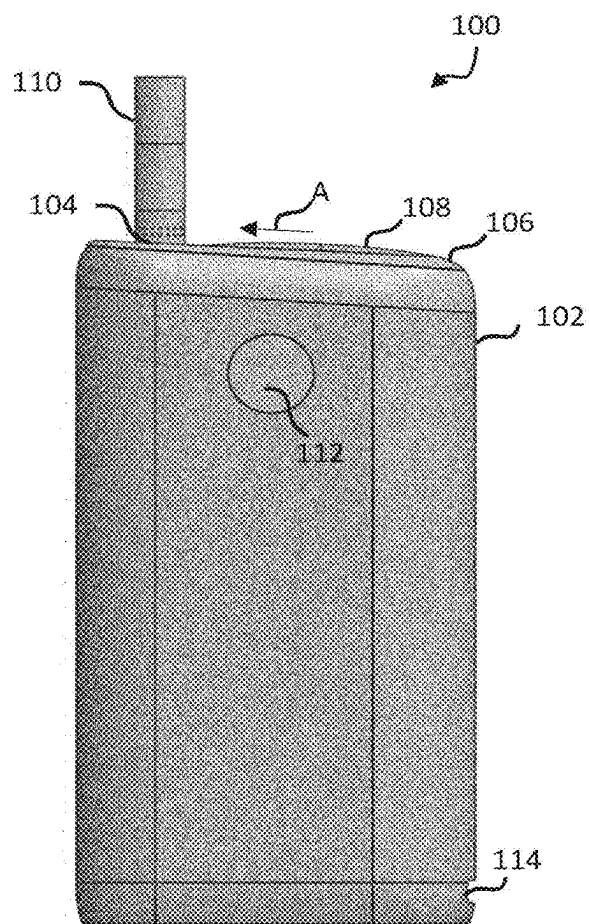
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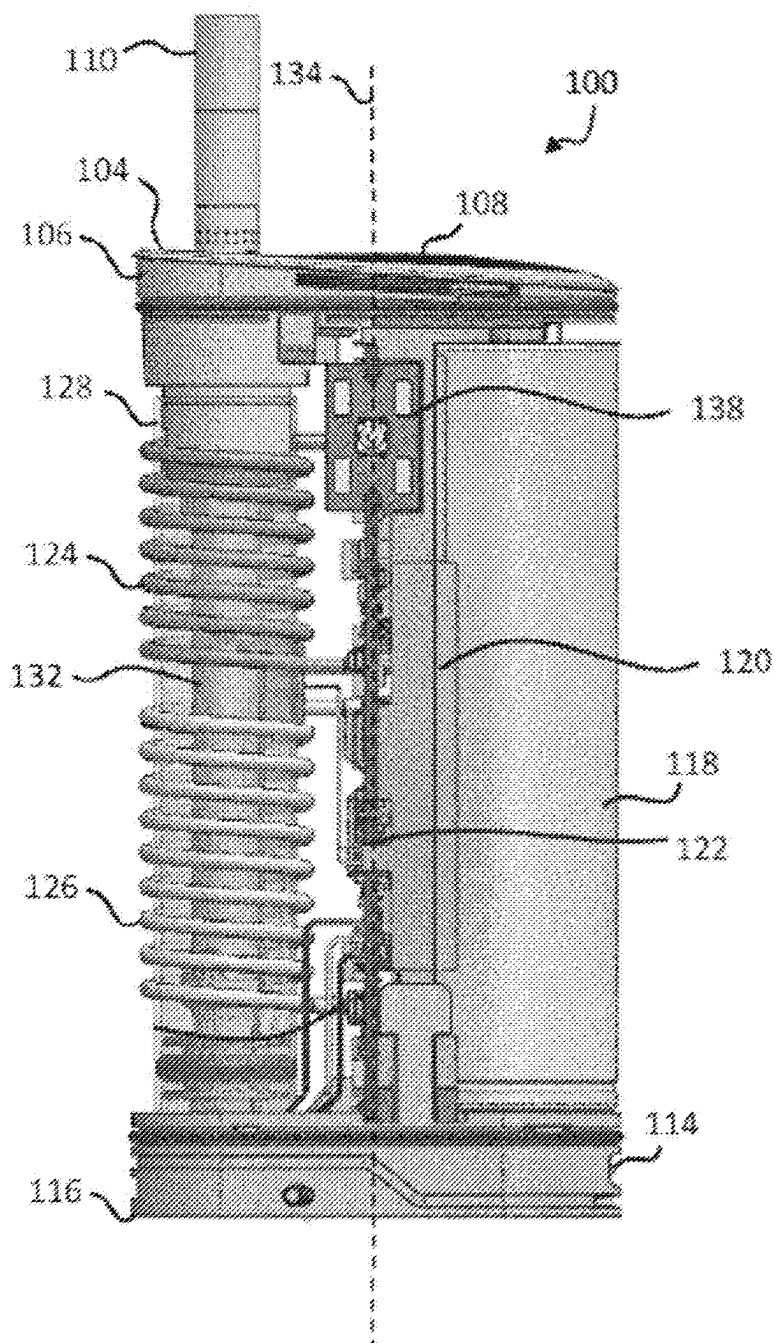
(57) **ABSTRACT**

Apparatus for an aerosol generating device is described. The apparatus comprises a heating circuit for heating a heating arrangement, the heating arrangement arranged to heat an aerosol generating material to thereby generate an aerosol; a temperature sensing arrangement for sensing a temperature of the device; and a controller for controlling a supply of energy to the heating circuit. The controller is configured to: determine a characteristic that is indicative that energy is being supplied to the heating circuit over a given time period, and determine a change in temperature sensed by the temperature sensing arrangement over the given time period; and take a control action if, based on the determined characteristic and the increase in temperature sensed by the temperature sensing arrangement over the given period, the controller determines that one or more predetermined criteria that are indicative of a fault with the temperature sensing arrangement are satisfied.

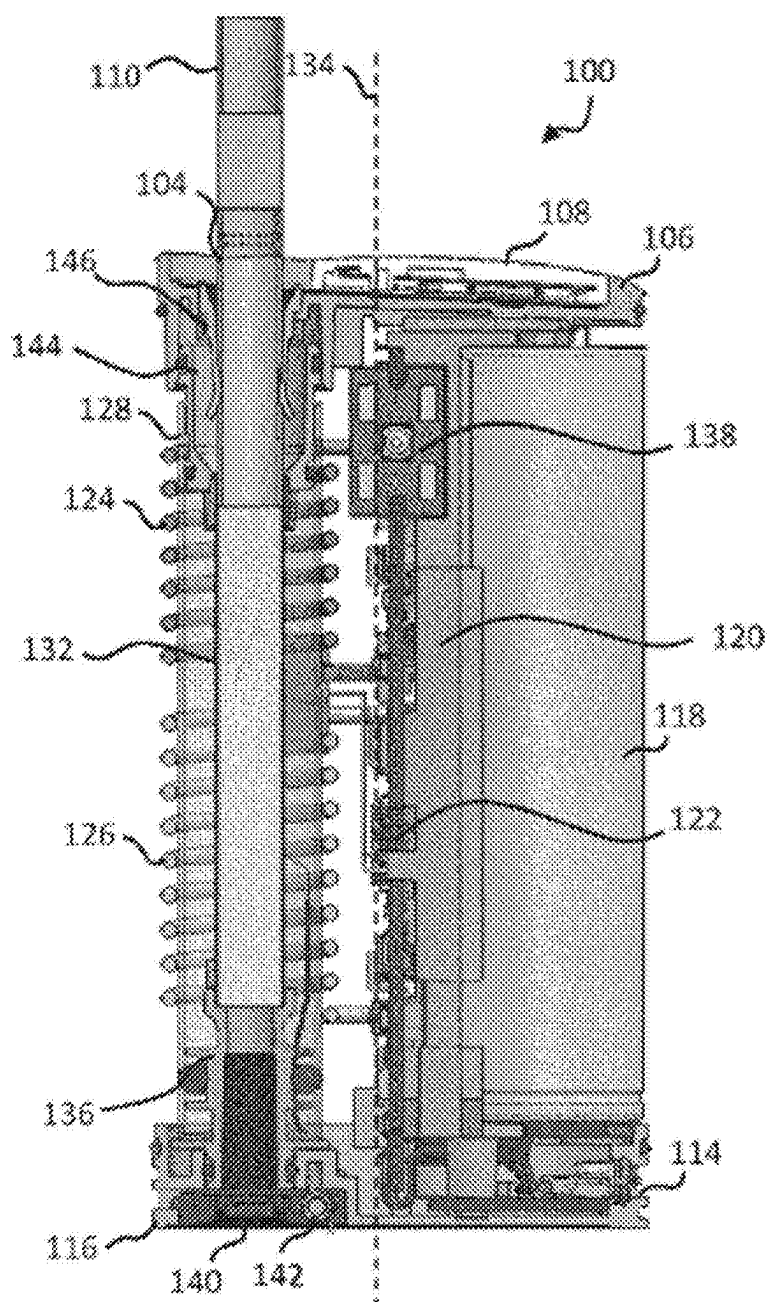




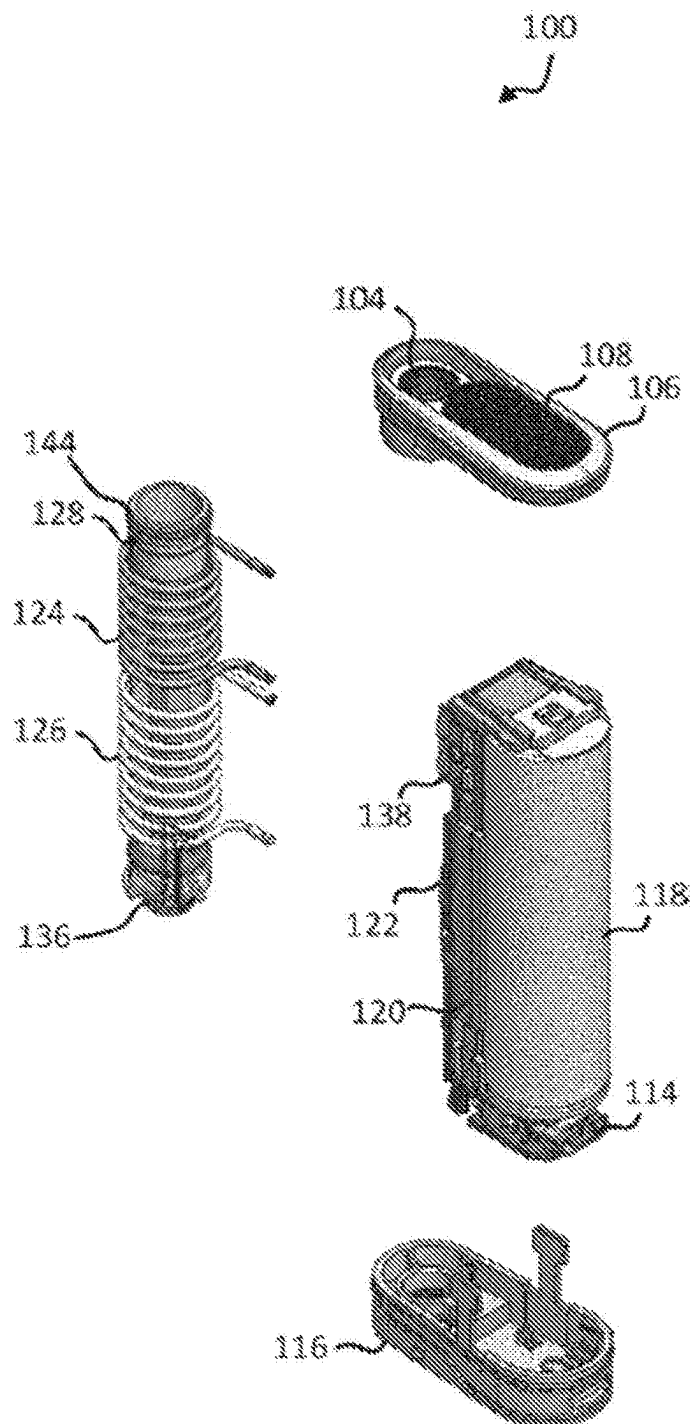
**Fig. 1**



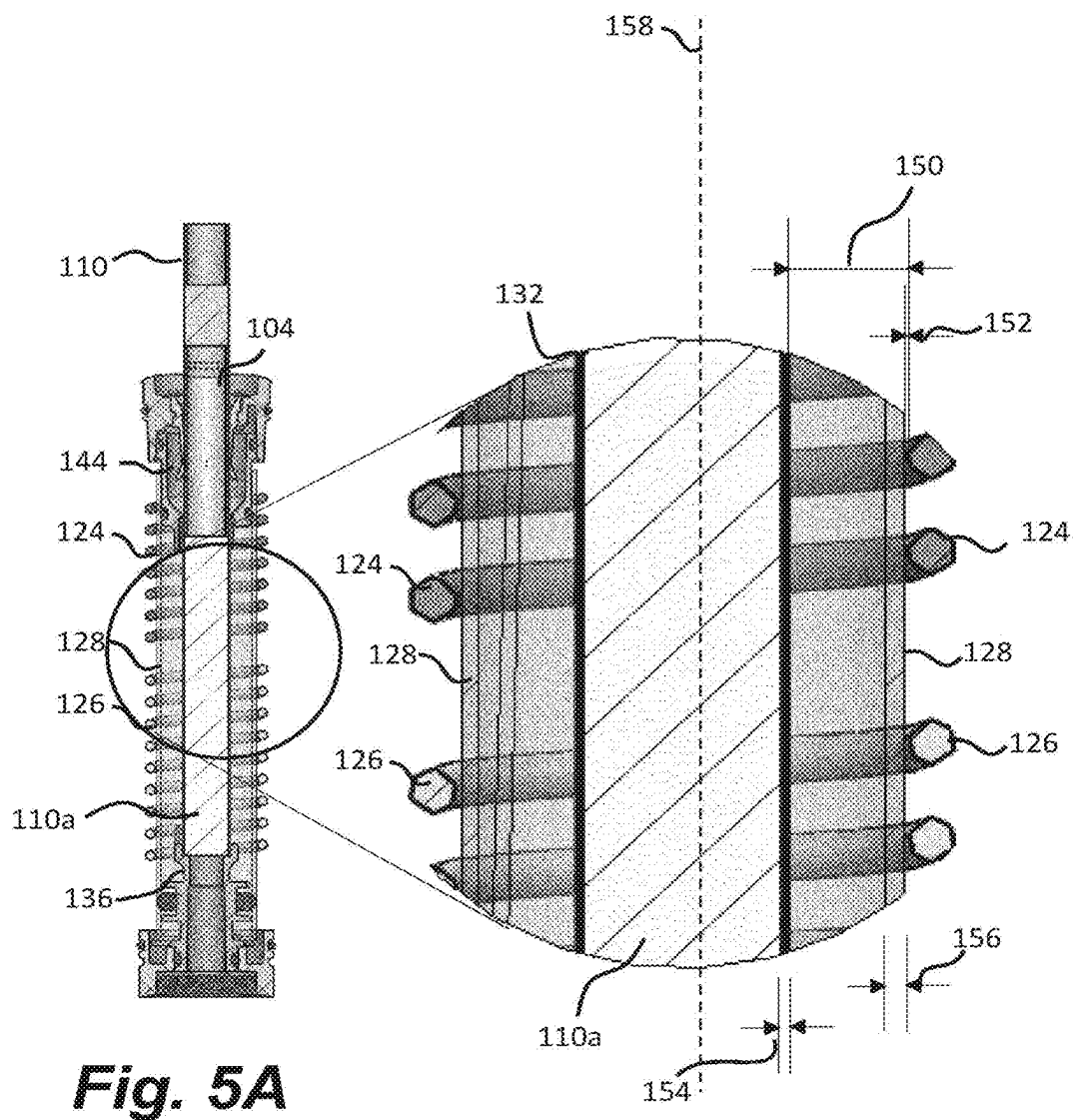
**Fig. 2**



**Fig. 3**



**Fig. 4**



**Fig. 5A**

**Fig. 5B**

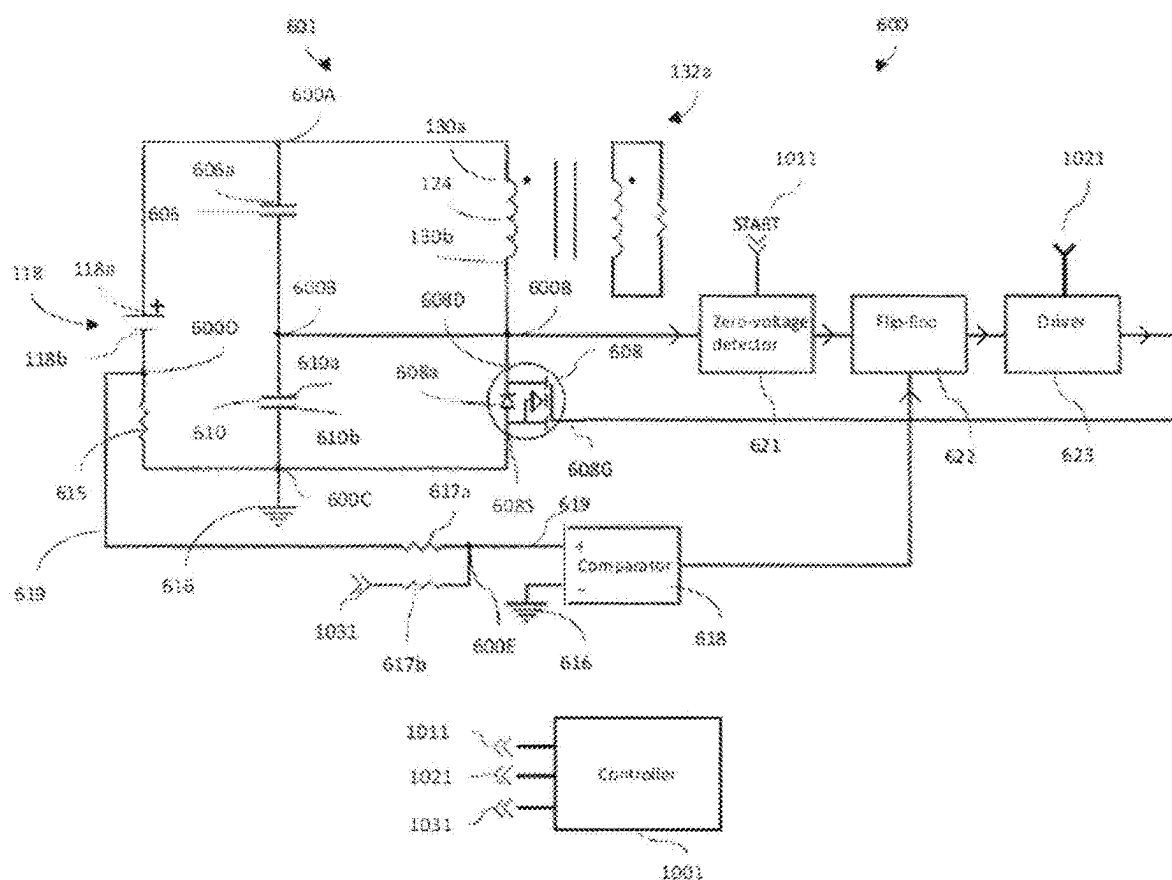
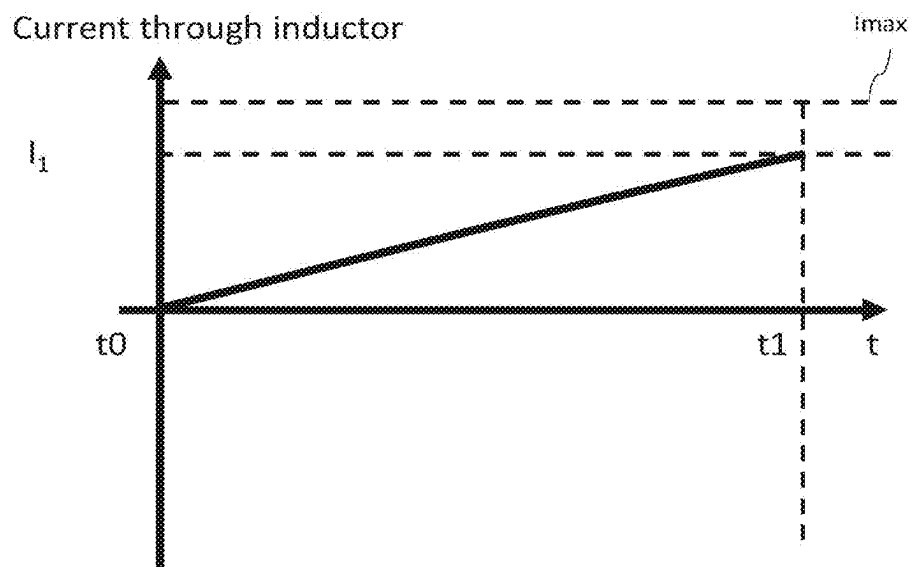
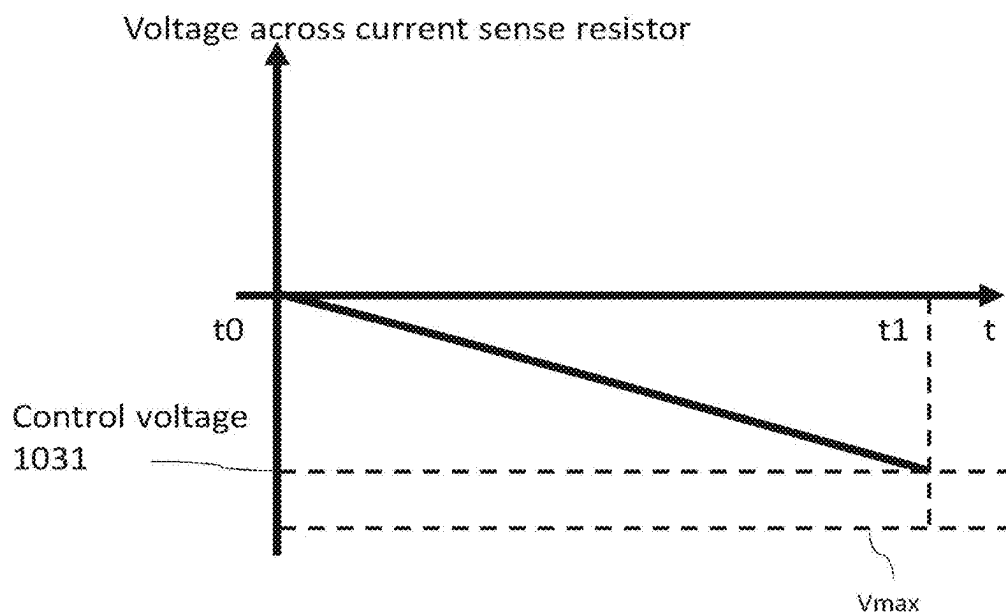


Fig. 6

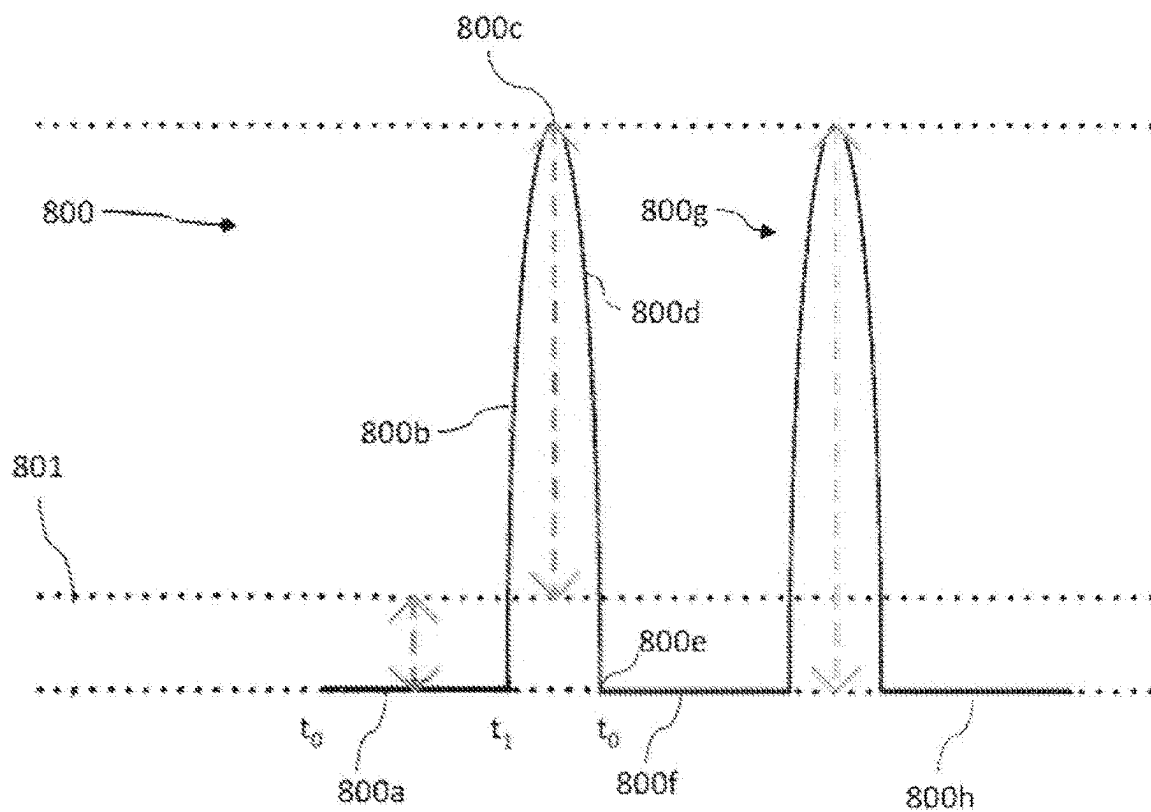


**Fig. 7A**

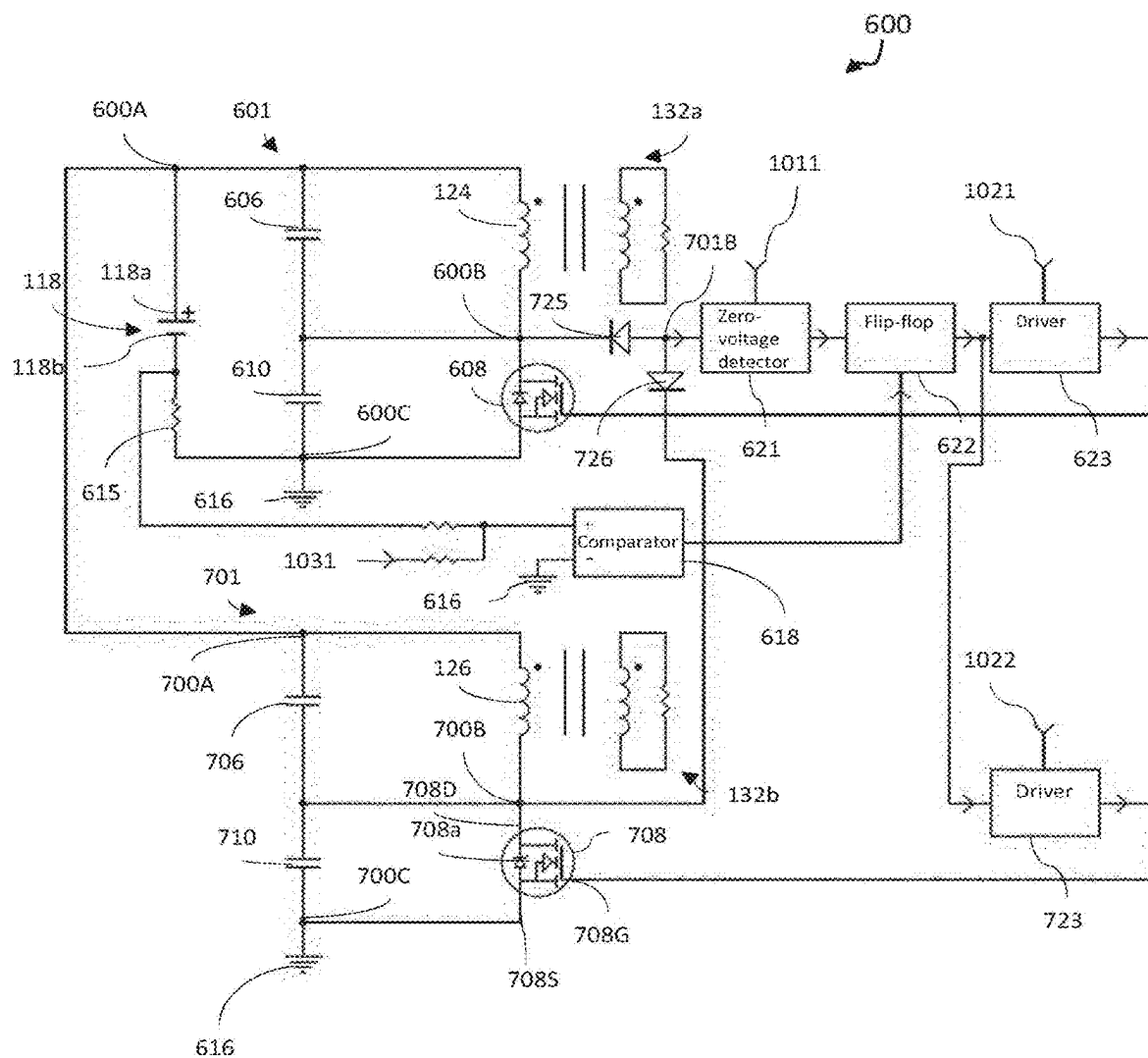


**Fig. 7B**

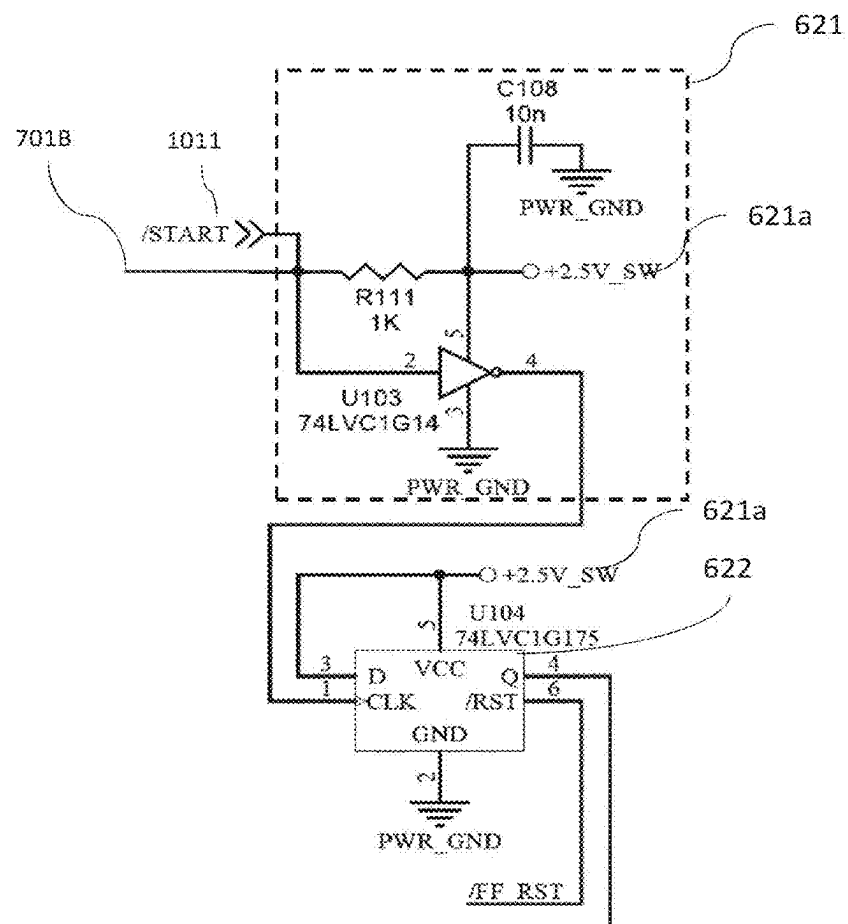




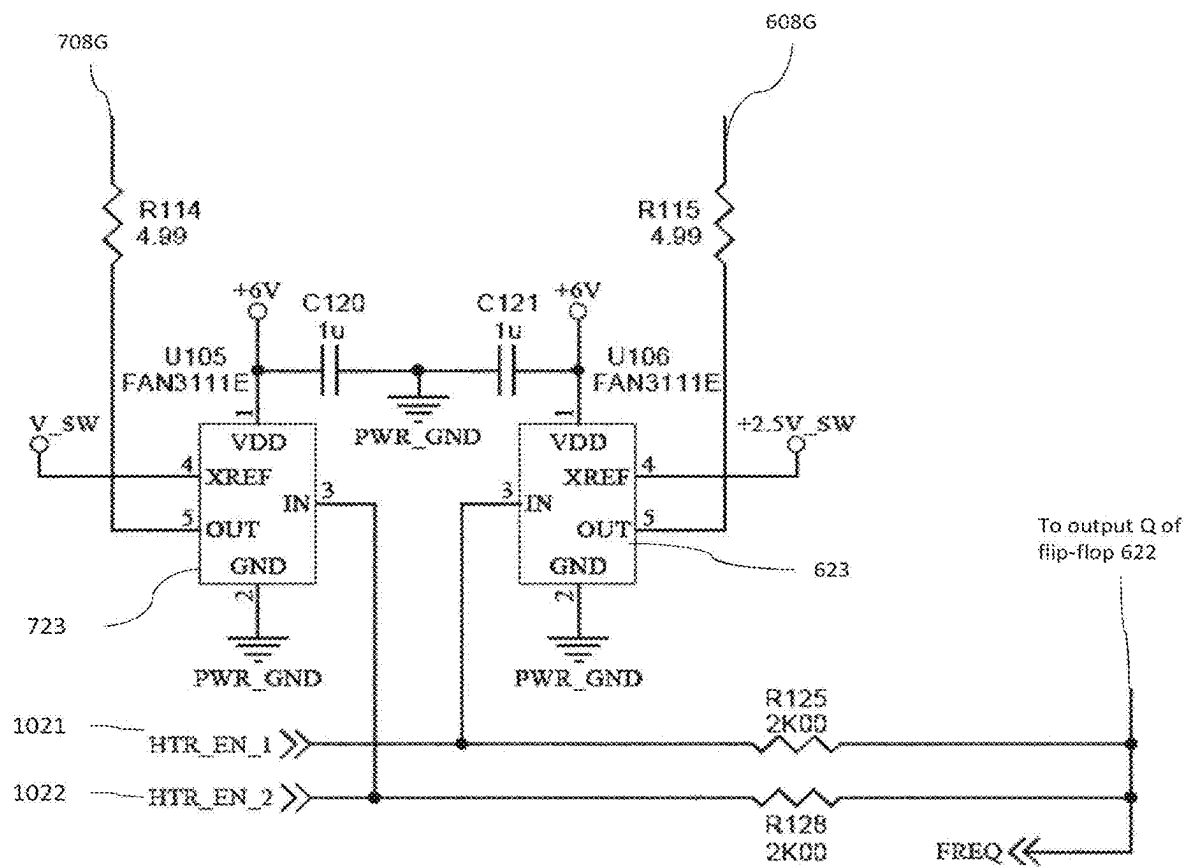
**Fig. 8**



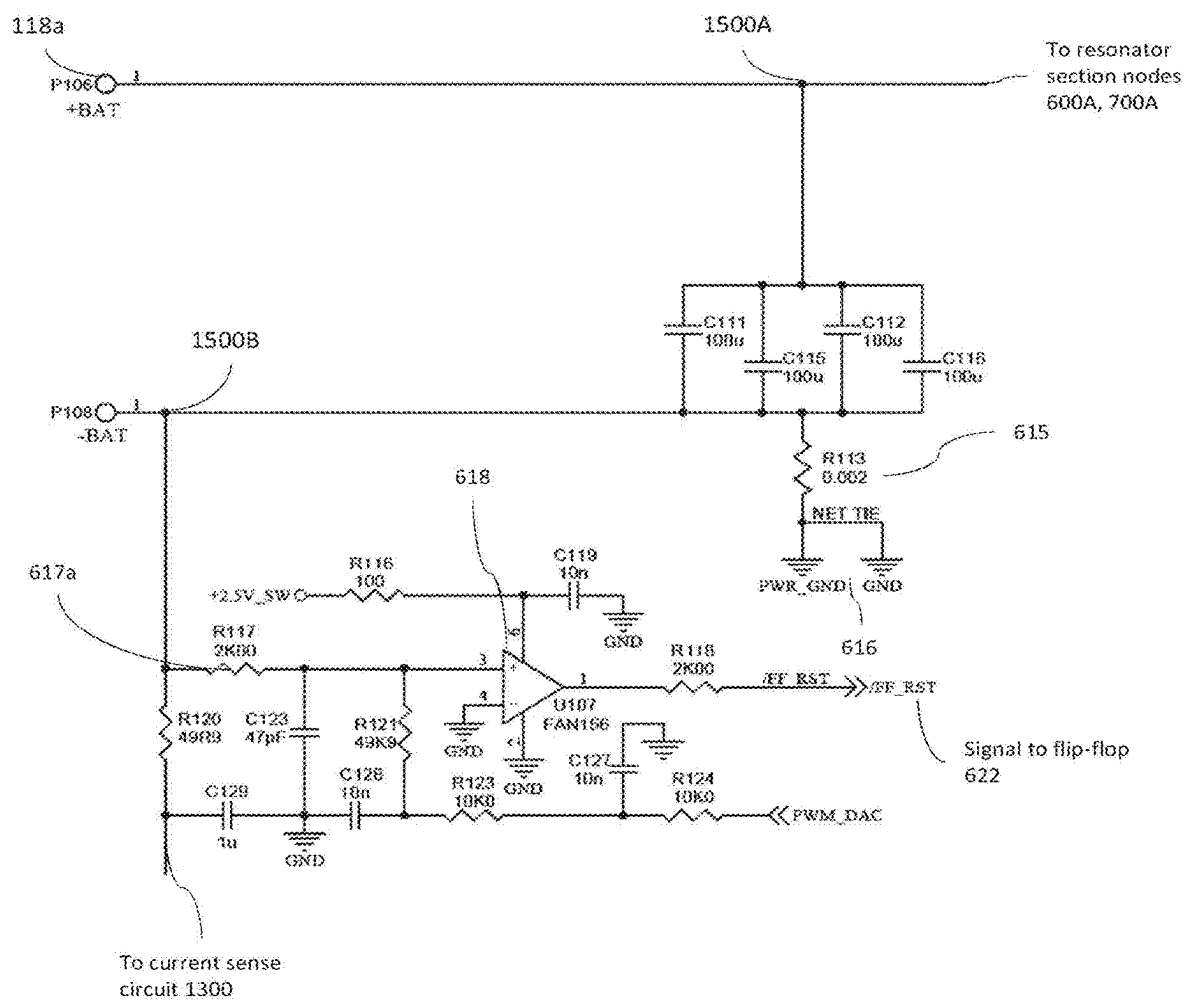
**Fig. 9**



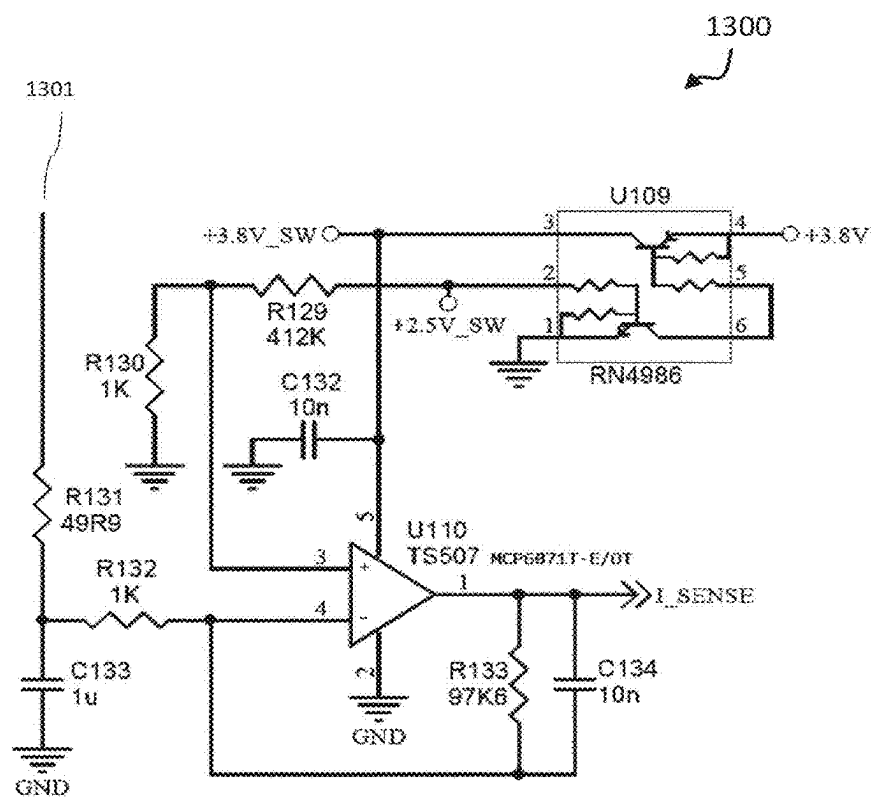
**Fig. 10**



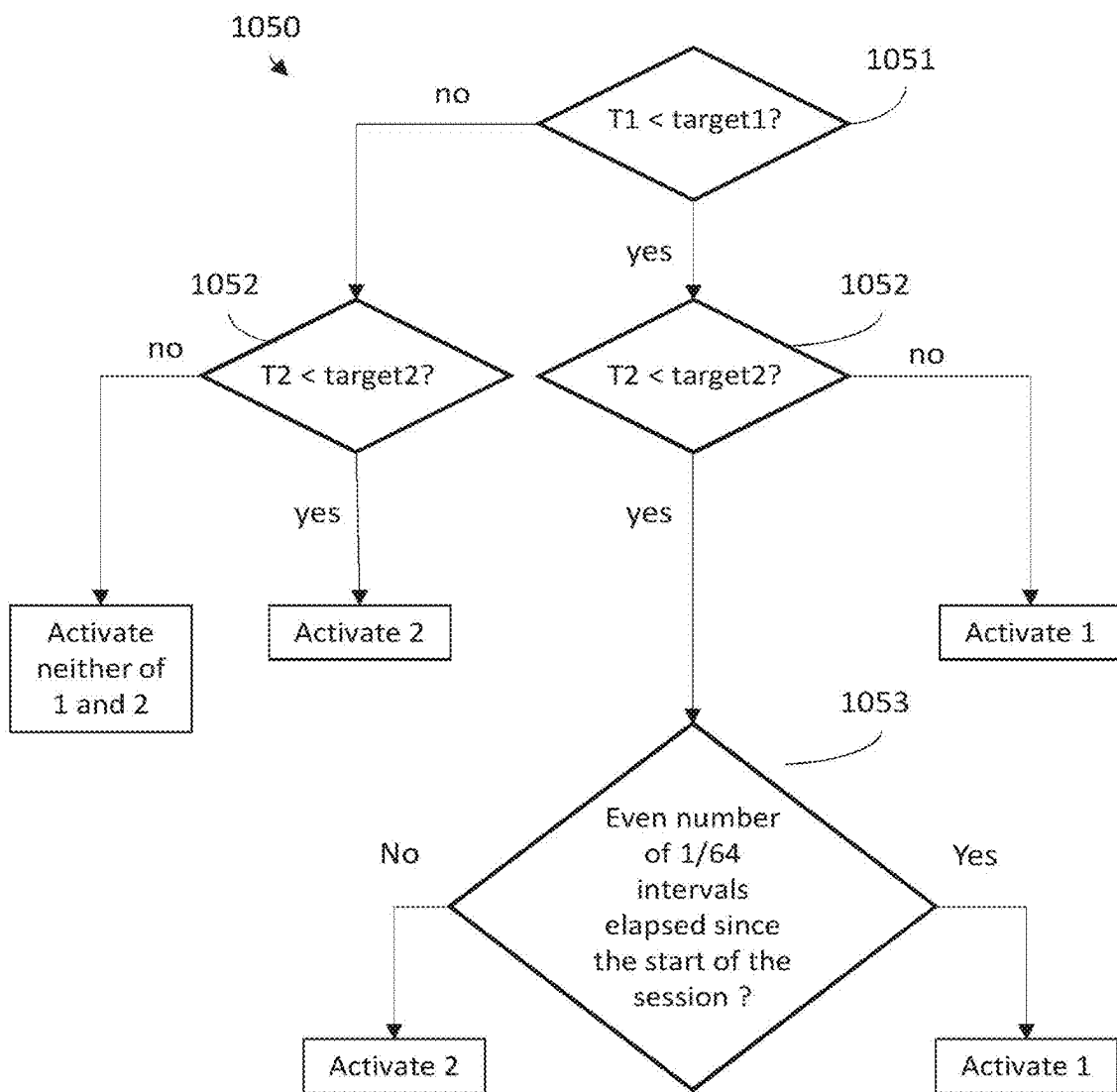
**Fig. 11**



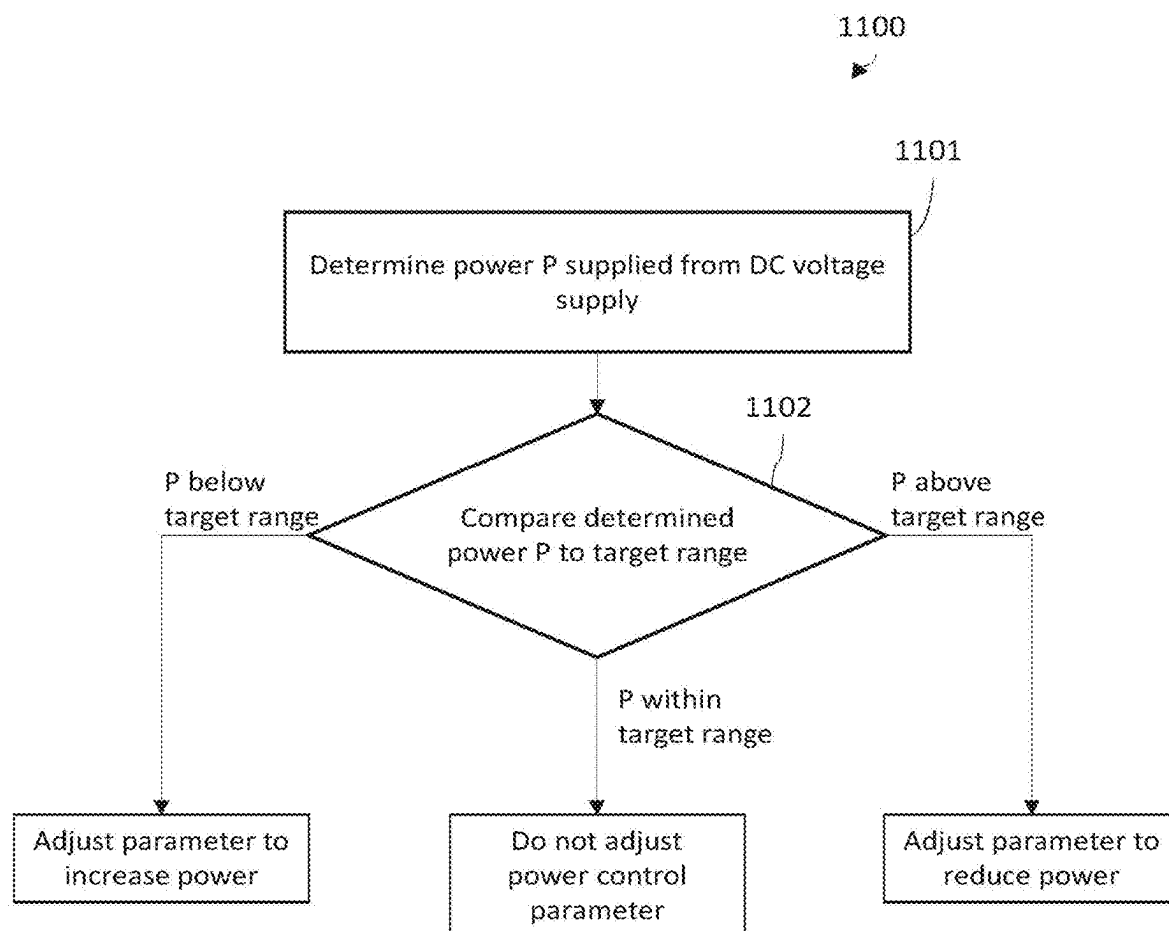
**Fig. 12**



**Fig. 13**

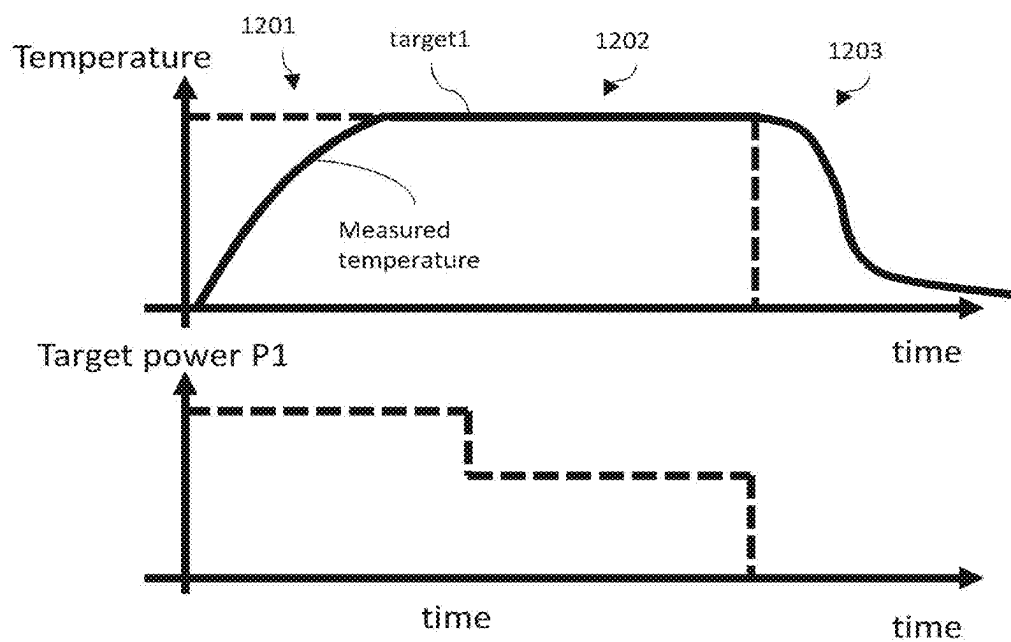


**Fig. 14**



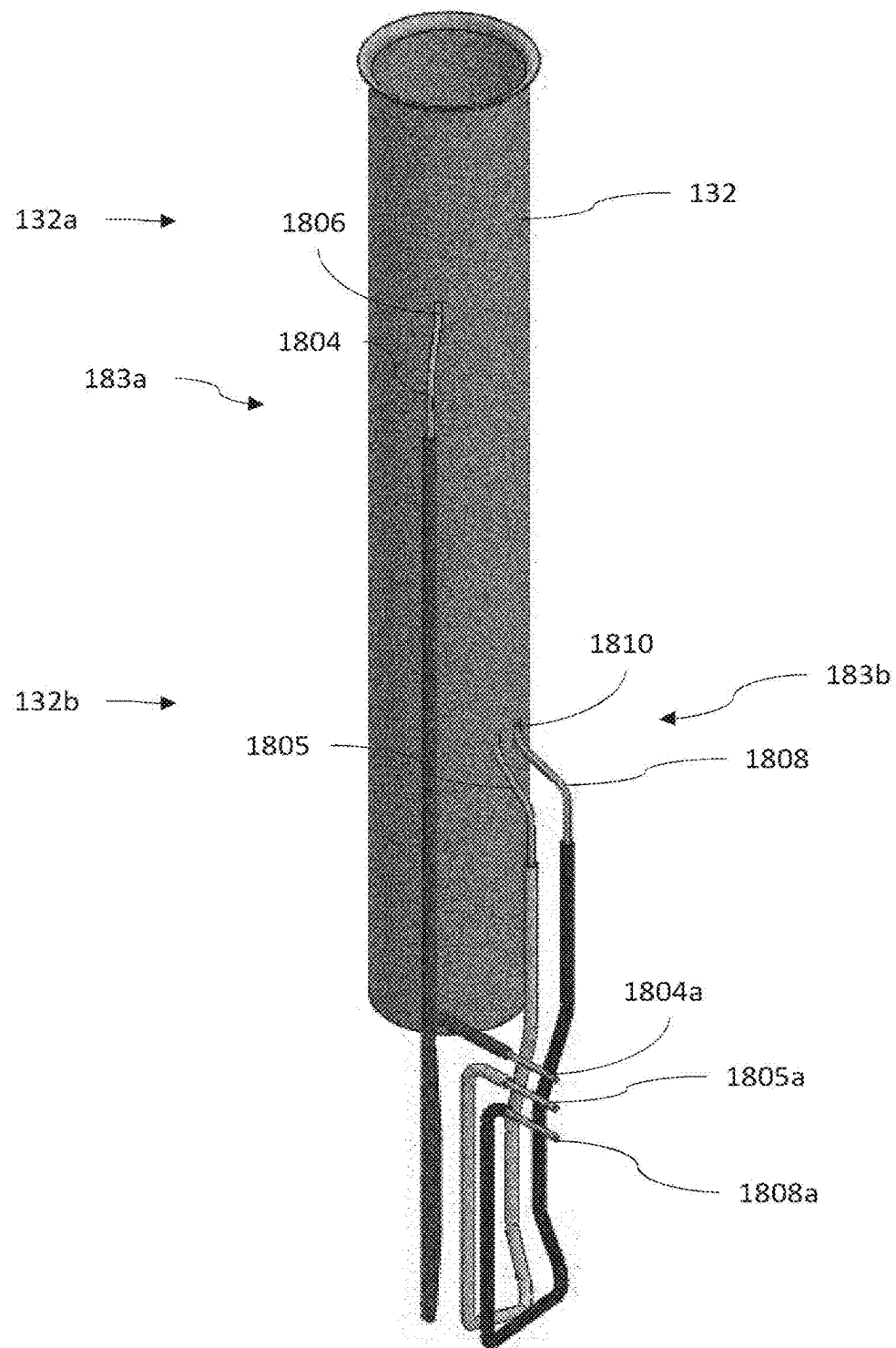
**Fig. 15**





**Fig. 16**





**Fig. 18**

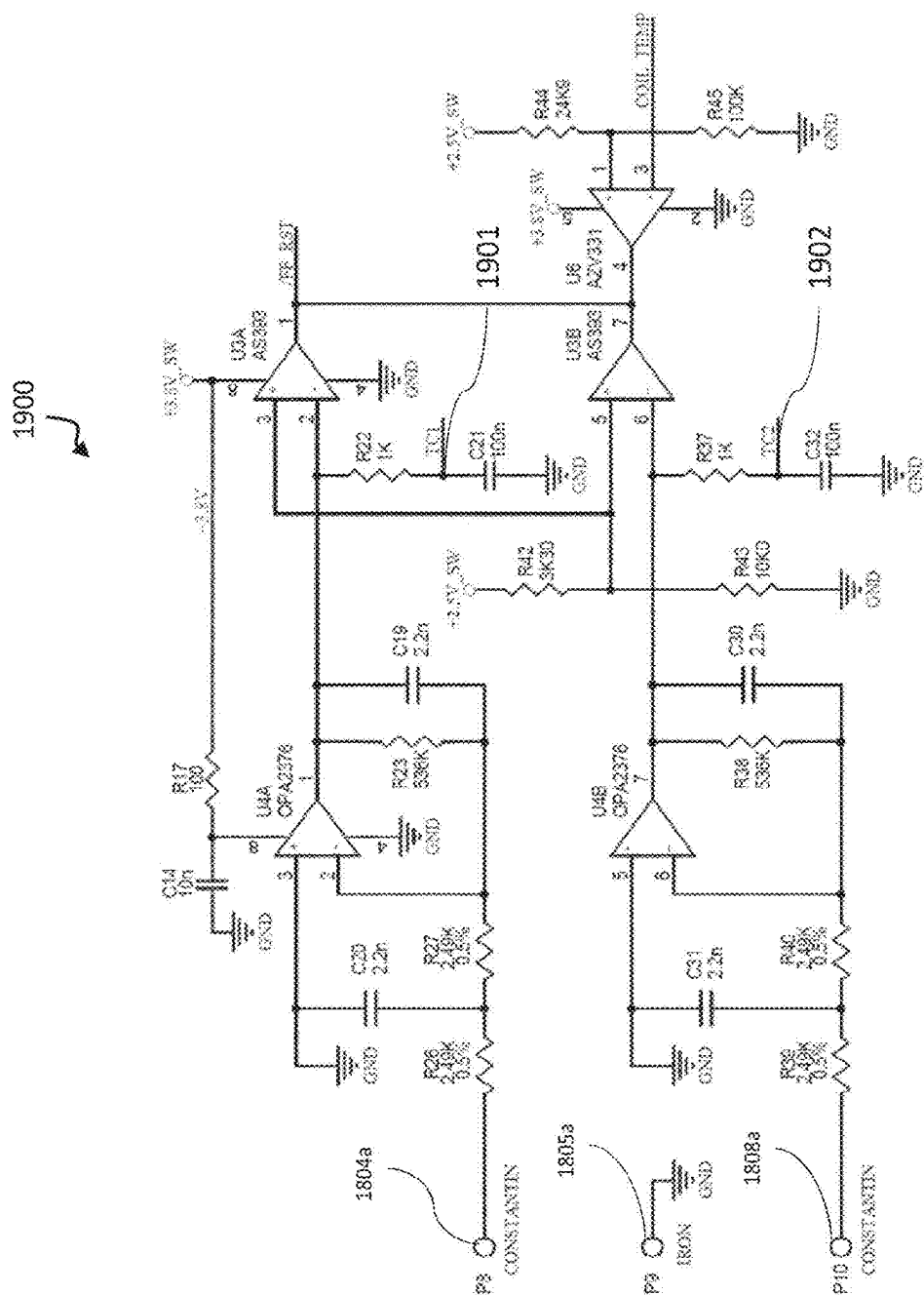
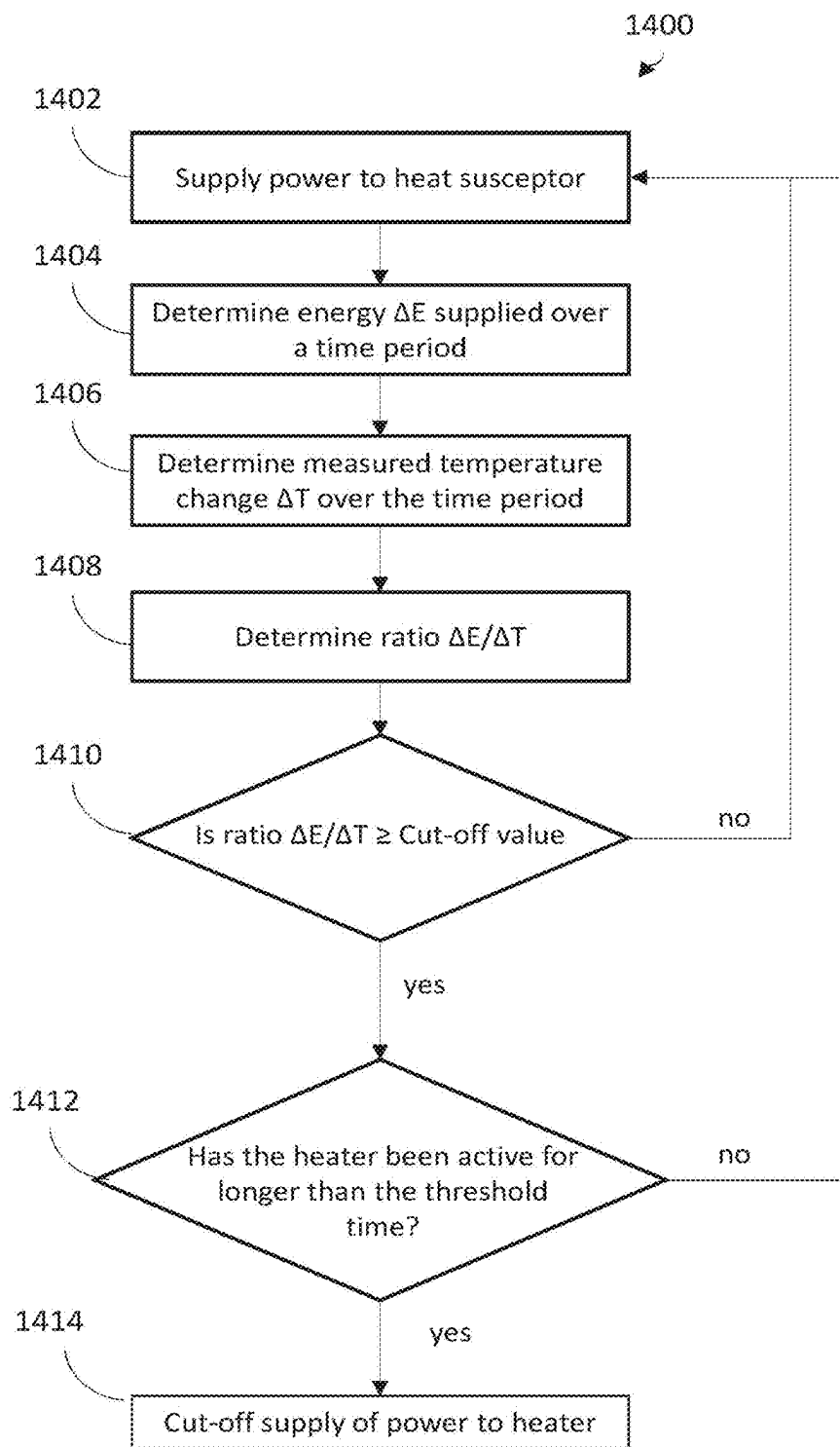


Fig. 19



**Fig. 20**

**APPARATUS FOR AEROSOL GENERATING  
DEVICE HAVING HEATING CONTROL  
CIRCUIT HEATING BASED ON  
TEMPERATURE OF THE HEATER OR  
HEATERS**

**PRIORITY CLAIM**

[0001] The present application is a continuation of U.S. application Ser. No. 17/593,159 filed Sep. 10, 2021, which is a national phase entry of International Application No. PCT/EP2020/056223, filed Mar. 9, 2020, which claims priority from U.S. provisional Application No. 62/816,287, filed Mar. 11, 2019, each of which is hereby fully incorporated by reference herein.

**TECHNICAL FIELD**

[0002] The present invention relates to apparatus for an aerosol generating device.

**BACKGROUND**

[0003] Smoking articles such as cigarettes, cigars, and the like, burn tobacco during use to create tobacco smoke. Attempts have been made to provide alternatives to these articles that burn tobacco by creating products that release compounds without burning. Examples of such products are heating devices which release compounds by heating, but not burning, the material. The material may be for example tobacco or other non-tobacco products, which may or may not contain nicotine.

**SUMMARY**

[0004] According to a first aspect of the present disclosure, there is provided apparatus for an aerosol generating device, the apparatus comprising: a heating circuit for heating a heating arrangement, the heating arrangement arranged to heat an aerosol generating material to thereby generate an aerosol; a temperature sensing arrangement for sensing a temperature of the device; and a controller for controlling a supply of energy to the heating circuit, wherein the controller is configured to: determine a characteristic that is indicative that energy is being supplied to the heating circuit over a given time period, and determine a change in temperature sensed by the temperature sensing arrangement over the given time period; and take a control action if, based on the determined characteristic and the increase in temperature sensed by the temperature sensing arrangement over the given period, the controller determines that one or more predetermined criteria that are indicative of a fault with the temperature sensing arrangement are satisfied.

[0005] The temperature sensing arrangement may comprise a temperature sensor for attaching to the heating element to sense a temperature of the heating element, and the one or more predetermined criteria are indicative that the temperature sensor is detached from the heating element.

[0006] The controller may be configured to determine a ratio of the amount of energy supplied to the heating circuit over the given period to the increase in temperature sensed by the temperature sensing arrangement over the given period; and take the control action if the supply of energy to the heating circuit if the ratio equals or exceeds a predetermined value.

[0007] The predetermined value for the ratio may be 2000 mJ/° C. to 6000 mJ/° C., or around 4000 mJ/° C.

[0008] The control action taken by the controller if the one or more predetermined criteria are satisfied may be to adjust the supply of energy to the heating circuit. For example, to stop the supply of energy to the heating circuit or to reduce the supply of energy to the heating circuit.

[0009] The controller may be configured to determine the predetermined criteria for the given time period and to determine the predetermined criteria once for each of one or more further predetermined periods in a usage session of the device, wherein, the predetermined periods may each have a duration of  $\frac{1}{80}$ s to  $\frac{1}{20}$ s or a duration of around  $\frac{1}{64}$ s.

[0010] The heating circuit may be an induction heating circuit, and the heating element may be a susceptor arrangement for being inductively heated by the induction heating circuit, and the temperature sensing arrangement may comprise a temperature sensor for sensing a temperature of the susceptor arrangement.

[0011] The temperature sensor may be a thermocouple for attaching to the susceptor arrangement.

[0012] The temperature sensing arrangement may comprise: a first temperature sensor for measuring a first temperature in the device; and a second temperature sensor for measuring a second temperature in the device; and the increase in temperature sensed by the temperature sensing arrangement over the given period may be an increase in the first temperature or an increase in the second temperature.

[0013] The first temperature may be a temperature of a first heating zone in the device and the second temperature may be a temperature of a second heating zone in the device.

[0014] The heating circuit may be configured to selectively heat the first heating zone and the second heating zone, and the controller may be configured to, during the given period, activate the heating circuit to heat only one of the first heating zone and the second heating zone.

[0015] The controller may be configured to determine the predetermined criteria for the given time period and to determine the predetermined criteria once for each of one or more further predetermined periods in a usage session of the device and during each period the heating circuit may be configured to selectively heat only one of the first heating zone and the second heating zone.

[0016] The increase in temperature sensed by the temperature sensing arrangement and used to determine the one or more criteria for each period during the usage session may be an increase in the first temperature if the heating circuit is active to heat the first heating zone during the period; and an increase in the second temperature if the heating circuit is active to heat the second heating zone during the period.

[0017] The heating circuit may be an induction heating circuit comprising a first inductor coil and a second inductor coil; and the heating element may be a susceptor arrangement, and the first heating zone may be a first zone of the susceptor arrangement arranged in use to be heated by the first inductor coil and the second heating zone may be a second zone of the susceptor arrangement arranged in use to be heated by the second inductor coil.

[0018] The first temperature sensor may be a first thermocouple for attaching to the first zone of the susceptor arrangement and the second temperature sensor may be a second thermocouple for attaching to the second zone of the susceptor arrangement.

[0019] The first thermocouple and second thermocouple may be J-type thermocouples each comprising a constantan wire and an iron wire.

[0020] The first thermocouple may comprise a first constantan wire and the second thermocouple may comprise a second constantan wire and the first thermocouple and second thermocouple may share a single iron wire.

[0021] According to a second aspect of the present disclosure there is provided an aerosol generating device comprising apparatus according to the first aspect of the present disclosure, wherein the aerosol provision device is for generating an aerosol for being inhaled by a user.

[0022] The device may be a tobacco heating device, also known as a heat-not-burn device.

[0023] According to a third aspect of the present disclosure there is provided an aerosol generating system comprising an aerosol generating device according to the second aspect and an article comprising an aerosol generating material for being heated by the device in use to thereby generate an aerosol.

[0024] Optionally, the aerosol generating material may comprise a tobacco material.

[0025] According to a fourth aspect of the present disclosure there is provided a method for a controller of apparatus for an aerosol generating device, the apparatus comprising: a heating circuit for heating a heating arrangement, the heating arrangement arranged to heat an aerosol generating material to thereby generate an aerosol; a temperature sensing arrangement for sensing a temperature of the device; and the controller, wherein the controller is for controlling a supply of energy to the heating circuit; wherein the method comprises: determining a characteristic that is indicative that energy is being supplied to the heating circuit during a given time period; determining a change in temperature sensed by the temperature sensing arrangement over the given time period; and taking a control action if, based on the characteristic and the increase in temperature sensed by the temperature sensing arrangement over the given period, the controller determines that one or more predetermined criteria that are indicative of a fault with the temperature sensing arrangement are satisfied.

[0026] According to a fifth aspect of the present disclosure there is provided a set of machine-readable instructions which when executed cause the method according to the fourth aspect to be performed.

[0027] According to a sixth aspect of the present disclosure there is provided a machine-readable medium comprising a set of instructions according to the fifth aspect.

[0028] Further features and advantages of the invention will become apparent from the following description of preferred embodiments of the invention, given by way of example only, which is made with reference to the accompanying drawings.

#### BRIEF DESCRIPTION OF THE DRAWINGS

[0029] Subject matter hereof may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying figures, in which:

[0030] FIG. 1 shows a front view of an example of an aerosol generating device;

[0031] FIG. 2 shows a front view of the aerosol generating device of FIG. 1 with an outer cover removed;

[0032] FIG. 3 shows a cross-sectional view of the aerosol generating device of FIG. 1;

[0033] FIG. 4 shows an exploded view of the aerosol generating device of FIG. 2;

[0034] FIG. 5A shows a cross-sectional view of a heating assembly within an aerosol generating device;

[0035] FIG. 5B shows a close-up view of a portion of the heating assembly of FIG. 5A;

[0036] FIG. 6 shows a schematic representation of an example induction heating circuit for the aerosol generating device of FIGS. 1 to 5B;

[0037] FIG. 7A shows a schematic representation of a current through an inductor of the example induction heating circuit of FIG. 6;

[0038] FIG. 7B shows a schematic representation of a voltage across a current sense resistor of the example induction heating circuit of FIG. 6;

[0039] FIG. 8 shows a schematic representation of a voltage across a switching arrangement of the circuit of FIG. 6;

[0040] FIG. 9 shows another schematic representation of the example induction heating circuit for the device of FIGS. 1 to 5B;

[0041] FIGS. 10 to 13 show various parts of an example control arrangement for the example induction heating circuit represented by previous figures;

[0042] FIG. 14 shows a flow chart representation of an example method of controlling aspects of an example induction heating circuit;

[0043] FIG. 15 shows a flow chart representation of another example method of controlling aspects of an example induction heating circuit;

[0044] FIG. 16 shows a schematic representation of a temperature of a susceptor and a target power to be supplied to heat the susceptor throughout an example usage session of an aerosol generating device;

[0045] FIG. 17 shows a perspective view of the susceptor and an example temperature sensing arrangement for measuring a temperature of the susceptor;

[0046] FIG. 18 shows a perspective view of the susceptor and another example temperature sensing arrangement for measuring the temperature of the susceptor;

[0047] FIG. 19 shows example apparatus for providing control functions in the device, where the control functions provided by the example apparatus of FIG. 19 relate to temperatures in the device; and

[0048] FIG. 20 shows a flowchart representation of a method of controlling power supplied to heat the susceptor.

#### DETAILED DESCRIPTION OF THE DRAWINGS

[0049] As used herein, the term “aerosol generating material” includes materials that provide volatilized components upon heating, typically in the form of an aerosol. Aerosol generating material includes any tobacco-containing material and may, for example, include one or more of tobacco, tobacco derivatives, expanded tobacco, reconstituted tobacco or tobacco substitutes. Aerosol generating material also may include other, non-tobacco, products, which, depending on the product, may or may not contain nicotine. Aerosol generating material may for example be in the form of a solid, a liquid, a gel, a wax or the like. Aerosol generating material may for example also be a combination or a blend of materials. Aerosol generating material may also be known as “smokable material”.

[0050] Apparatuses are known that heat aerosol generating material to volatilize at least one component of the aerosol generating material, typically to form an aerosol which can be inhaled, without burning or combusting the

aerosol generating material. Such an apparatus is sometimes described as an “aerosol generating device,” an “aerosol provision device,” a “heat-not-burn device,” a “tobacco heating product device,” or a “tobacco heating device” or similar. Similarly, there are also so-called e-cigarette devices, which typically vaporize an aerosol generating material in the form of a liquid, which may or may not contain nicotine. The aerosol generating material may be in the form of or be provided as part of a rod, cartridge or cassette or the like which can be inserted into the apparatus. A heater for heating and volatilizing the aerosol generating material may be provided as a “permanent” part of the apparatus.

[0051] An aerosol provision device can receive an article comprising aerosol generating material for heating. An “article” in this context is a component that includes or contains in use the aerosol generating material, which is heated to volatilize the aerosol generating material, and optionally other components in use. A user may insert the article into the aerosol provision device before it is heated to produce an aerosol, which the user subsequently inhales. The article may be, for example, of a predetermined or specific size that is configured to be placed within a heating chamber of the device which is sized to receive the article.

[0052] FIG. 1 shows an example of an aerosol provision device 100 for generating aerosol from an aerosol generating medium/material. In broad outline, the device 100 may be used to heat a replaceable article 110 comprising the aerosol generating medium, to generate an aerosol or other inhalable medium which is inhaled by a user of the device 100.

[0053] The device 100 comprises a housing 102 (in the form of an outer cover) which surrounds and houses various components of the device 100. The device 100 has an opening 104 in one end, through which the article 110 may be inserted for heating by a heating assembly. In use, the article 110 may be fully or partially inserted into the heating assembly where it may be heated by one or more components of the heater assembly.

[0054] The device 100 of this example comprises a first end member 106 which comprises a lid 108 which is moveable relative to the first end member 106 to close the opening 104 when no article 110 is in place. In FIG. 1, the lid 108 is shown in an open configuration, however the cap 108 may move into a closed configuration. For example, a user may cause the lid 108 to slide in the direction of arrow “A”.

[0055] The device 100 may also include a user-operable control element 112, such as a button or switch, which operates the device 100 when pressed. For example, a user may turn on the device 100 by operating the switch 112.

[0056] The device 100 may also comprise an electrical component, such as a socket/port 114, which can receive a cable to charge a battery of the device 100. For example, the socket 114 may be a charging port, such as a USB charging port. In some examples the socket 114 may be used additionally or alternatively to transfer data between the device 100 and another device, such as a computing device.

[0057] FIG. 2 depicts the device 100 of FIG. 1 with the outer cover 102 removed. The device 100 defines a longitudinal axis 134.

[0058] As shown in FIG. 2, the first end member 106 is arranged at one end of the device 100 and a second end member 116 is arranged at an opposite end of the device 100. The first and second end members 106, 116 together at least

partially define end surfaces of the device 100. For example, the bottom surface of the second end member 116 at least partially defines a bottom surface of the device 100. Edges of the outer cover 102 may also define a portion of the end surfaces. In this example, the lid 108 also defines a portion of a top surface of the device 100. FIG. 2 also shows a second printed circuit board 138 associated within the control element 112.

[0059] The end of the device closest to the opening 104 may be known as the proximal end (or mouth end) of the device 100 because, in use, it is closest to the mouth of the user. In use, a user inserts an article 110 into the opening 104, operates the user control 112 to begin heating the aerosol generating material and draws on the aerosol generated in the device. This causes the aerosol to flow through the device 100 along a flow path towards the proximal end of the device 100.

[0060] The other end of the device furthest away from the opening 104 may be known as the distal end of the device 100 because, in use, it is the end furthest away from the mouth of the user. As a user draws on the aerosol generated in the device, the aerosol flows away from the distal end of the device 100.

[0061] The device 100 further comprises a power source 118. The power source 118 may be, for example, a battery, such as a rechargeable battery or a non-rechargeable battery. Examples of suitable batteries include, for example, a lithium battery (such as a lithium-ion battery), a nickel battery (such as a nickel-cadmium battery), and an alkaline battery. The battery is electrically coupled to the heating assembly to supply electrical power when required and under control of a controller (not shown in FIG. 2) to heat the aerosol generating material. In this example, the battery is connected to a central support 120 which holds the battery 118 in place.

[0062] The device further comprises at least one electronics module 122. The electronics module 122 may comprise, for example, a printed circuit board (PCB). The PCB 122 may support at least one controller, such as a processor, and memory. The PCB 122 may also comprise one or more electrical tracks to electrically connect together various electronic components of the device 100. For example, the battery terminals may be electrically connected to the PCB 122 so that power can be distributed throughout the device 100. The socket 114 may also be electrically coupled to the battery via the electrical tracks.

[0063] In the example device 100, the heating assembly is an inductive heating assembly and comprises various components to heat the aerosol generating material of the article 110 via an inductive heating process. Induction heating is a process of heating an electrically conducting object (such as a susceptor) by electromagnetic induction. An induction heating assembly may comprise an inductive element, for example, one or more inductor coils, and a device for passing a varying electric current, such as an alternating electric current, through the inductive element. The varying electric current in the inductive element produces a varying magnetic field. The varying magnetic field penetrates a susceptor suitably positioned with respect to the inductive element, and generates eddy currents inside the susceptor. The susceptor has electrical resistance to the eddy currents, and hence the flow of the eddy currents against this resistance causes the susceptor to be heated by Joule heating. In cases where the susceptor comprises ferromagnetic material



such as iron, nickel or cobalt, heat may also be generated by magnetic hysteresis losses in the susceptor, i.e. by the varying orientation of magnetic dipoles in the magnetic material as a result of their alignment with the varying magnetic field. In inductive heating, as compared to heating by conduction for example, heat is generated inside the susceptor, allowing for rapid heating. Further, there need not be any physical contact between the inductive heater and the susceptor, allowing for enhanced freedom in construction and application.

**[0064]** The induction heating assembly of the example device **100** comprises a susceptor arrangement **132** (herein referred to as “a susceptor”), a first inductor coil **124** and a second inductor coil **126**. The first and second inductor coils **124**, **126** are made from an electrically conducting material. In this example, the first and second inductor coils **124**, **126** are made from litz wire/cable which is wound in a helical fashion to provide helical inductor coils **124**, **126**. Litz wire comprises a plurality of individual wires which are individually insulated and are twisted together to form a single wire. Litz wires are designed to reduce the skin effect losses in a conductor. In the example device **100**, the first and second inductor coils **124**, **126** are made from copper litz wire which has a substantially circular cross section. In other examples the litz wire can have other shape cross sections, such as rectangular.

**[0065]** The first inductor coil **124** is configured to generate a first varying magnetic field for heating a first section of the susceptor **132** and the second inductor coil **126** is configured to generate a second varying magnetic field for heating a second section of the susceptor **132**. Herein, the first section of the susceptor **132** is referred to as the first susceptor zone **132a** and the second section of the susceptor **132** is referred to as the second susceptor zone **132b**. In this example, the first inductor coil **124** is adjacent to the second inductor coil **126** in a direction along the longitudinal axis **134** of the device **100** (that is, the first and second inductor coils **124**, **126** do not overlap). In this example the susceptor arrangement **132** comprises a single susceptor comprising two zones, however in other examples the susceptor arrangement **132** may comprise two or more separate susceptors. In some examples, there may be more than two heating zones. Each zone may be formed by respective parts of a single susceptor of the susceptor arrangement or by separate susceptors of the susceptor arrangement. Ends **130** of the first and second inductor coils **124**, **126** are connected to the PCB **122**.

**[0066]** It will be appreciated that the first and second inductor coils **124**, **126**, in some examples, may have at least one characteristic different from each other. For example, the first inductor coil **124** may have at least one characteristic different from the second inductor coil **126**. More specifically, in one example, the first inductor coil **124** may have a different value of inductance than the second inductor coil **126**. In FIG. 2, the first and second inductor coils **124**, **126** are of different lengths such that the first inductor coil **124** is wound over a smaller section of the susceptor **132** than the second inductor coil **126**. Thus, the first inductor coil **124** may comprise a different number of turns than the second inductor coil **126** (assuming that the spacing between individual turns is substantially the same). In yet another example, the first inductor coil **124** may be made from a different material to the second inductor coil **126**. In some examples, the first and second inductor coils **124**, **126** may be substantially identical.

**[0067]** In this example, the inductor coils **124**, **126** are wound in the same direction as one another. That is, both the first inductor coil **124**, and the second inductor coil **126** are left-hand helices. In another example, both inductor coils **124**, **126** may be right-hand helices. In yet another example (not shown), the first inductor coil **124** and the second inductor coil **126** are wound in opposite directions. This can be useful when the inductor coils are active at different times. For example, initially, the first inductor coil **124** may be operating to heat a first section of the article **110**, and at a later time, the second inductor coil **126** may be operating to heat a second section of the article **110**. Winding the coils in opposite directions helps reduce the current induced in the inactive coil when used in conjunction with a particular type of control circuit. In one example where the coils **124**, **126** are wound in different directions (not shown) the first inductor coil **124** may be a right-hand helix and the second inductor coil **126** may be a left-hand helix. In another such embodiment, the first inductor coil **124** may be a left-hand helix and the second inductor coil **126** may be a right-hand helix.

**[0068]** The susceptor **132** of this example is hollow and therefore defines a receptacle within which aerosol generating material is received. For example, the article **110** can be inserted into the susceptor **132**. In this example the susceptor **132** is tubular, with a circular cross section.

**[0069]** The device **100** of FIG. 2 further comprises an insulating member **128** which may be generally tubular and at least partially surround the susceptor **132**. The insulating member **128** may be constructed from any insulating material, such as a plastics material for example. In this particular example, the insulating member is constructed from polyether ether ketone (PEEK). The insulating member **128** may help insulate the various components of the device **100** from the heat generated in the susceptor **132**.

**[0070]** The insulating member **128** can also fully or partially support the first and second inductor coils **124**, **126**. For example, as shown in FIG. 2, the first and second inductor coils **124**, **126** are positioned around the insulating member **128** and are in contact with a radially outward surface of the insulating member **128**. In some examples the insulating member **128** does not abut the first and second inductor coils **124**, **126**. For example, a small gap may be present between the outer surface of the insulating member **128** and the inner surface of the first and second inductor coils **124**, **126**.

**[0071]** In a specific example, the susceptor **132**, the insulating member **128**, and the first and second inductor coils **124**, **126** are coaxial around a central longitudinal axis of the susceptor **132**.

**[0072]** FIG. 3 shows a side view of device **100** in partial cross-section. The outer cover **102** is again not present in this example. The circular cross-sectional shape of the first and second inductor coils **124**, **126** is more clearly visible in FIG. 3.

**[0073]** The device **100** further comprises a support **136** which engages one end of the susceptor **132** to hold the susceptor **132** in place. The support **136** is connected to the second end member **116**.

**[0074]** The device **100** further comprises a second lid/cap **140** and a spring **142**, arranged towards the distal end of the device **100**. The spring **142** allows the second lid **140** to be

opened, to provide access to the susceptor 132. A user may, for example, open the second lid 140 to clean the susceptor 132 and/or the support 136.

[0075] The device 100 further comprises an expansion chamber 144 which extends away from a proximal end of the susceptor 132 towards the opening 104 of the device. Located at least partially within the expansion chamber 144 is a retention clip 146 to abut and hold the article 110 when received within the device 100. The expansion chamber 144 is connected to the end member 106.

[0076] FIG. 4 is an exploded view of the device 100 of FIG. 1, with the outer cover 102 again omitted.

[0077] FIG. 5A depicts a cross section of a portion of the device 100 of FIG. 1. FIG. 5B depicts a close-up of a region of FIG. 5A. FIGS. 5A and 5B show the article 110 received within the susceptor 132, where the article 110 is dimensioned so that the outer surface of the article 110 abuts the inner surface of the susceptor 132. This ensures that the heating is most efficient. The article 110 of this example comprises aerosol generating material 110a. The aerosol generating material 110a is positioned within the susceptor 132. The article 110 may also comprise other components such as a filter, wrapping materials and/or a cooling structure.

[0078] FIG. 5B shows that the outer surface of the susceptor 132 is spaced apart from the inner surface of the inductor coils 124, 126 by a distance 150, measured in a direction perpendicular to a longitudinal axis 158 of the susceptor 132. In one particular example, the distance 150 is about 3 mm to 4 mm, about 3 mm to 3.5 mm, or about 3.25 mm.

[0079] FIG. 5B further shows that the outer surface of the insulating member 128 is spaced apart from the inner surface of the inductor coils 124, 126 by a distance 152, measured in a direction perpendicular to a longitudinal axis 158 of the susceptor 132. In one particular example, the distance 152 is about 0.05 mm. In another example, the distance 152 is substantially 0 mm, such that the inductor coils 124, 126 abut and touch the insulating member 128.

[0080] In one example, the susceptor 132 has a wall thickness 154 of about 0.025 mm to 1 mm, or about 0.05 mm.

[0081] In one example, the susceptor 132 has a length of about 40 mm to 60 mm, about 40 mm to 45 mm, or about 44.5 mm.

[0082] In one example, the insulating member 128 has a wall thickness 156 of about 0.25 mm to 2 mm, 0.25 mm to 1 mm, or about 0.5 mm.

[0083] As has been described above, the heating assembly of the example device 100 is an inductive heating assembly comprising various components to heat the aerosol generating material of article 110 via an induction heating process. In particular, the first inductor coil 124 and the second inductor coil 126 are used to heat respective first 132a and second 132b zones of the susceptor 132 in order to heat the aerosol generating material and generate an aerosol. Below, with reference to FIGS. 6 to 12, the operation of the device 100 in using the first and second inductor coils 124, 126 to inductively heat the susceptor arrangement 132 will be described in detail.

[0084] The inductive heating assembly of the device 100 comprises an LC circuit. An LC circuit, has an inductance L provided by an induction element, and a capacitance C provided by a capacitor. In the device 100, the inductance L

is provided by the first and second inductor coils 124, 126 and the capacitance C is provided by a plurality of capacitors as will be discussed below. An induction heater circuit comprising an inductance L and a capacitance C may in some cases be represented as an RLC circuit, comprising a resistance R provided by a resistor. In some cases, resistance is provided by the ohmic resistance of parts of the circuit connecting the inductor and the capacitor, and hence the circuit need not necessarily include a resistor as such. Such circuits may exhibit electrical resonance, which occurs at a particular resonant frequency when the imaginary parts of impedances or admittances of circuit elements cancel each other.

[0085] One example of an LC circuit is a series circuit where the inductor and capacitor are connected in series. Another example of an LC circuit is a parallel LC circuit where the inductor and capacitor are connected in parallel. Resonance occurs in an LC circuit because the collapsing magnetic field of the inductor generates an electric current in its windings that charges the capacitor, while the discharging capacitor provides an electric current that builds the magnetic field in the inductor. When a parallel LC circuit is driven at the resonant frequency, the dynamic impedance of the circuit is at maximum (as the reactance of the inductor equals the reactance of the capacitor), and circuit current is at a minimum. However, for a parallel LC circuit, the parallel inductor and capacitor loop acts as a current multiplier (effectively multiplying the current within the loop and thus the current passing through the inductor). Allowing the RLC or LC circuit to operate at the resonant frequency for at least some of the time while the circuit is in operation to heat the susceptor may therefore provide for effective and/or efficient inductive heating by providing for the greatest value of the magnetic field penetrating the susceptor.

[0086] The LC circuit used by the device 100 to heat the susceptor 132 may make use of one or more transistors acting as a switching arrangement as will be described below. A transistor is a semiconductor device for switching electronic signals. A transistor typically comprises at least three terminals for connection to an electronic circuit. A field effect transistor (FET) is a transistor in which the effect of an applied electric field may be used to vary the effective conductance of the transistor. The field effect transistor may comprise a body, a source terminal S, a drain terminal D, and a gate terminal G. The field effect transistor comprises an active channel comprising a semiconductor through which charge carriers, electrons or holes, may flow between the source S and the drain D. The conductivity of the channel, i.e. the conductivity between the drain D and the source S terminals, is a function of the potential difference between the gate G and source S terminals, for example generated by a potential applied to the gate terminal G. In enhancement mode FETs, the FET may be OFF (i.e. substantially prevent current from passing therethrough) when there is substantially zero gate G to source S voltage, and may be turned ON (i.e. substantially allow current to pass therethrough) when there is a substantially non-zero gate G-source S voltage.

[0087] One type of transistor which may be used in circuitry of the device 100 is an n-channel (or n-type) field effect transistor (n-FET). An n-FET is a field effect transistor whose channel comprises an n-type semiconductor, where electrons are the majority carriers and holes are the minority carriers. For example, n-type semiconductors may comprise an intrinsic semiconductor (such as silicon for example)

doped with donor impurities (such as phosphorus for example). In n-channel FETs, the drain terminal D is placed at a higher potential than the source terminal S (i.e. there is a positive drain-source voltage, or in other words a negative source-drain voltage). In order to turn an n-channel FET “on” (i.e. to allow current to pass therethrough), a switching potential is applied to the gate terminal G that is higher than the potential at the source terminal S.

[0088] Another type of transistor which may be used in the device 100 is a p-channel (or p-type) field effect transistor (p-FET). A p-FET is a field effect transistor whose channel comprises a p-type semiconductor, where holes are the majority carriers and electrons are the minority carriers. For example, p-type semiconductors may comprise an intrinsic semiconductor (such as silicon for example) doped with acceptor impurities (such as boron for example). In p-channel FETs, the source terminal S is placed at a higher potential than the drain terminal D (i.e. there is a negative drain-source voltage, or in other words a positive source-drain voltage). In order to turn a p-channel FET “on” (i.e. to allow current to pass therethrough), a switching potential is applied to the gate terminal G that is lower than the potential at the source terminal S (and which may for example be higher than the potential at the drain terminal D).

[0089] In examples, one or more of the FETs used in the device 100 may be a metal-oxide-semiconductor field effect transistor (MOSFET). A MOSFET is a field effect transistor whose gate terminal G is electrically insulated from the semiconductor channel by an insulating layer. In some examples, the gate terminal G may be metal, and the insulating layer may be an oxide (such as silicon dioxide for example), hence “metal-oxide-semiconductor”. However, in other examples, the gate may be made from other materials than metal, such as polysilicon, and/or the insulating layer may be made from other materials than oxide, such as other dielectric materials. Such devices are nonetheless typically referred to as metal-oxide-semiconductor field effect transistors (MOSFETs), and it is to be understood that as used herein the term metal-oxide-semiconductor field effect transistors or MOSFETs is to be interpreted as including such devices.

[0090] A MOSFET may be an n-channel (or n-type) MOSFET where the semiconductor is n-type. The n-channel MOSFET (n-MOSFET) may be operated in the same way as described above for the n-channel FET. As another example, a MOSFET may be a p-channel (or p-type) MOSFET, where the semiconductor is p-type. The p-channel MOSFET (p-MOSFET) may be operated in the same way as described above for the p-channel FET. An n-MOSFET typically has a lower source-drain resistance than that of a p-MOSFET. Hence in an “on” state (i.e. where current is passing therethrough), n-MOSFETs generate less heat as compared to p-MOSFETs, and hence may waste less energy in operation than p-MOSFETs. Further, n-MOSFETs typically have shorter switching times (i.e. a characteristic response time from changing the switching potential provided to the gate terminal G to the MOSFET changing whether or not current passes therethrough) as compared to p-MOSFETs. This can allow for higher switching rates and improved switching control.

[0091] Now with reference to FIG. 6, circuitry for induction heating by the device 100 will be described. FIG. 6 shows a schematic representation of a part of an induction heating circuit 600 of the aerosol generating device 100.

FIG. 6 shows a part of the induction heating circuit 600 which comprises the first inductor coil 124 for heating the first susceptor zone 132a when a varying current flows through the first inductor coil 124. The first susceptor zone 132a is represented in FIG. 6 as having an inductive element and a resistive element to represent how the susceptor 132 couples inductively with the first inductor 124 and is heated through the generation of eddy currents. It will be noted that the device 100 additionally comprises the second inductor coil 126, which is not shown in FIG. 6. The second inductor coil 126 is also part of the induction heating circuit 600 and is controlled to heat the second susceptor zone 132b as will be described below. However, for the sake of clarity, the circuit 600 will first be described with reference to those features shown in FIG. 6.

[0092] The circuit 600 comprises a first resonator section 601, the DC voltage supply 118 for supplying a DC voltage to the first resonator section 601, as well as a control arrangement for controlling the circuit 600. The first resonator section 601 comprises the first inductor 124 and a switching arrangement comprising a first FET 608, and the control arrangement is configured to switch the FET 608 between a first state and a second state in response to voltage conditions detected in the circuit 600, as will be described in more detail below, to operate the first inductor 124. The circuit 600, with the exception of the susceptor 132, is arranged on the PCB 122 of the device 100, with the inductor coil 124 being connected to the PCB 122 at a first end 130a and a second end 130b.

[0093] The first resonator section 601 comprises a first capacitor 606, and a second capacitor 610, both arranged in parallel with the first inductor 124 such that when the first resonator section 601 is allowed to resonate an alternating current flows between the first capacitor 606 and the second capacitor 610 and through the inductor 124. As mentioned above, the first FET 608, in this example an n-channel MOSFET, is arranged to operate as a switching arrangement in the first resonator section 601.

[0094] It should be noted that in other examples, the resonator section 601 may comprise only one capacitor, for example in the position of the first capacitor 606, or at the position of the second capacitor 610. In other examples, the resonator section 601 may comprise any other number of capacitors, such as three or more capacitors. For example, either or both of the first capacitor 606 and the second capacitor 610 may be replaced by two or more capacitors arranged in parallel with one another. As will be well understood, the resonator section 601 has a resonant frequency which is dependent on the inductance L and the capacitance C of the resonator section 601. The number, type and arrangement of capacitors in the resonating section 601 may be selected based on considerations of the power levels to be used in the circuit 600 and the desired frequency of operation of the circuit 600. For example, it will be understood that individual capacitors and an arrangement of said capacitors can be considered to have an equivalent series resistance (ESR) as well as a limit on the ability of said capacitors to handle current. Such features may be taken into account when determining an arrangement of capacitors to provide the capacitance in the resonator section 601. For example, depending on desired power levels and frequency of operation, there may be an advantage to providing a plurality of capacitors in parallel, to provide higher capacitance or lower ESR. In this example, the first and second

capacitors **606**, **610** are both ceramic COG capacitors each having a capacitance of around 100 nF. In other examples, other types of capacitors and/or capacitors with other capacitance values, e.g. capacitors with unequal capacitance values, may be used, according to the considerations outlined in this paragraph.

[0095] The first resonator section **601** is supplied a DC voltage by the DC voltage supply **118**, which is, for example, as described above, a voltage supplied by a battery. As shown in FIG. 6, the DC voltage supply **118** comprises a positive terminal **118a** and a negative terminal **118b**. In one example, the DC voltage supply **118** supplies a DC voltage of around 4.2V to the first resonator section **601**. In other examples, the DC voltage supply **118** may supply a voltage of 2 to 10V, or around 3 to 5V, for example.

[0096] A controller **1001** is configured to control operation of the circuit **600**. The controller **1001** may comprise a micro-controller, e.g. a micro-processing unit (MPU), comprising a plurality of inputs and outputs. In one example, the controller **1001** is an STM32L051C8T6 model MPU. In some examples, the DC voltage supply **118** provided to the circuit **600** is provided by an output from the controller **1001** which itself receives power from a battery or other power source.

[0097] The positive terminal **118a** of the DC voltage source **118** is electrically connected to a first node **600A**. In an example, the DC voltage source **118** is connected to the node **600A** via the controller **1001** which receives power from the DC voltage source **118** and supplies the voltage supplied by the DC voltage source to components of the device, including the circuit **600**. The first node **600A** is electrically connected to a first end **606a** of the first capacitor **606** and to the first end **130a** of the first inductor **124**. The second end **130b** of the first inductor **124** is electrically connected to a second node **600B**, which in FIG. 6 is represented at two electrically equivalent points in the circuit diagram. The second node **600B** is electrically connected to a drain terminal **608D** of the FET **608**. In this example, the second node **600B** is also electrically connected to a first end **610a** of the second capacitor **610**. Continuing around the circuit, the source terminal **608S** of the first FET **608** is electrically connected to a third node **600C**. The third node **600C** is electrically connected to ground **616**, and in this example to a second end **610b** of the second capacitor **610**. The third node **600C** is electrically connected via a current sense resistor **615** to a fourth node **600D**, and the fourth node **600D** is electrically connected to the negative terminal **118b** of the DC voltage source **118**, which, as with the positive terminal, in an example is supplied via the controller **1001**.

[0098] It should be noted that in examples where the second capacitor **610** is not present, the third node **600C** may have only three electrical connections: to the first FET source terminal **608S**, to ground **616** and to the current sense resistor **615**.

[0099] As mentioned above, the first FET **608** acts a switching arrangement in the first resonator section **601**. The first FET **608** is configurable between a first state, i.e. an 'ON' state and a second state, i.e. an 'OFF' state. As will be well understood by those skilled in the art when an n-channel FET is in an OFF state (i.e. when the appropriate control voltage is not applied to its gate) it effectively acts as a diode. In FIG. 6, the diode functionality that the first FET **608** exhibits when in its OFF state is represented by a first

diode **608a**. That is, when the FET **608** is in the OFF state the first diode **608a** acts to largely prevent current flowing from the drain terminal **608D** to the source terminal **608S** but allows current to flow from the source terminal **608S** to the drain terminal **608D** if the diode **608a** is appropriately forward biased. An n-channel FET is in an ON state when an appropriate control voltage is applied to its gate so that a conductive path exists between its drain D and source S. As such, when the first FET **608** is in the ON state, it acts like a closed switch in the first resonator section **601**.

[0100] As mentioned above, the circuit **600** may be considered to comprise a first resonator section **601** and an additional control arrangement. The control arrangement comprises a comparator **618**, a zero-voltage detector **621**, and a flip-flop **622**, and is configured to detect voltage conditions within the first resonator section **601** and to control the first FET **608** in response to the detected voltage conditions. This control of the first FET **608** by the control arrangement will now be described in more detail.

[0101] At the second node **600B** there is electrically connected the zero-voltage detector **621**, which is configured to detect a voltage condition, i.e. a voltage of at or near 0V with respect to a ground voltage, at a point in the circuit **600** to which the zero-voltage detector **621** is connected. The zero-voltage detector **621** is configured to output a signal to control switching of the state of the FET **608**. That is, the zero-voltage detector **621** is configured to output a signal to the flip-flop **622**. The flip-flop **622** is an electrical circuit which is configurable between two stable states. The flip-flop **622** is electrically connected to a first gate driver **623** which is configured to provide a voltage to the first FET gate terminal **608G** dependent on the state of the flip-flop. That is, the first gate driver **623** is configured to provide an appropriate voltage to the first FET gate terminal **608G** to switch the FET **608** to the ON state when the flip-flop is in one state, but is configured not to provide a voltage appropriate for maintaining the FET **608** in the ON state when the flip-flop **622** is in the other state. For example, the first gate driver **623** may be configured to provide an appropriate gate-source voltage to the first FET gate **608G** to switch the FET **608** ON when the flip-flop **622** is in a state '1', and the first gate driver **623** may be configured not to provide the gate-source voltage when the flip-flop **622** is in state '0'. The state of the flip-flop means **622** therefore controls whether the first FET **608** is on or off.

[0102] In this example, the zero-voltage detector **621** and the first gate driver **623** of the control arrangement are configured to receive respective signals **1011**, **1021** from the controller **1001**, by which signals the controller **1001** can initiate and control operation of the circuit **600**, as will be discussed in more detail below.

[0103] At the fourth node **600D**, there is electrically connected a control voltage line **619**. The control voltage line **619** is electrically connected to a fifth node **600E** via a resistor **617a** and the fifth node **600E** is electrically connected to the voltage comparator **618**—hereinafter comparator **618**. The fifth node **600E** is electrically connected to a positive terminal of the comparator **618**. A negative terminal of the comparator **618** is connected to ground **616**. In this example, the comparator **618** is configured to output a signal based on a comparison of the voltage at the fifth node **600E** to ground voltage. The output signal of the comparator **618** is sent to the flip-flop **622**. A control voltage **1031** is

supplied, in this example from the controller 1001, to the control voltage line 619 via a second resistor 617b.

[0104] As mentioned above, the comparator 618 is electrically connected to provide an output to the flip-flop 622. The flip-flop 622 is configured such that an output signal from the comparator 618 can change the state of the flip-flop 622, and thereby cause the first driver 623 to change the state of the first FET 608.

[0105] The functioning of the example circuit 600 will now be described in more detail in the context of the first resonator section 601 being activated by the controller 1001 such that the first inductor coil 124 is operated to heat the first susceptor zone 132a.

[0106] To begin, the first FET 608 is configured in the OFF state, and is thus acting as a diode 608a, preventing current flowing through the inductor 124. The controller 1001 initiates the operation of the circuit 600 to heat the first susceptor zone 132a by causing the FET 608 to switch from the OFF state to the ON state. In this example the controller initiates operation of the circuit 600 by providing a START signal 1011 to the zero-voltage detector 621. The flip-flop 622 is thereby caused to change states and cause the first gate driver 623 to provide a signal to the FET gate terminal 608G to thereby switch the FET to the ON state.

[0107] Once the FET 608 is switched to the ON state, what may be referred to as a self-oscillating heating cycle of the circuit 600 begins. The FET 608, now being in the ON state, acts as a closed switch allowing a DC current to begin flowing from the DC voltage source positive terminal 118a through the first inductor 124 and returning to the DC voltage source negative terminal 118b via the current sense resistor 615. The first inductor 124 opposes this initial increase in current, as is well-known, generating a back electromotive force (EMF) as understood via Faraday's and Lenz's laws. In the ON state, the voltage between the drain terminal 608D and the source terminal 608S is substantially zero.

[0108] FIG. 7A shows a schematic graphical representation of the current flowing through the first inductor 124 against time  $t$  starting from when the FET 608 is switched on, at time  $t_0$ . From time  $t_0$ , a DC current begins to build up in the inductor 124 from zero at a rate which is dependent on an inductance  $L_1$  of the inductor 124, a DC resistance of the circuit 600 and the DC supply voltage. In one example the current sense resistor 615 has a resistance of around 2 m $\Omega$ , while the inductor 124 has a DC resistance of, 2 to 15 m $\Omega$ , or 4 to 10 m $\Omega$  or in this example around 5.2 m $\Omega$ . This build-up of current in the inductor corresponds to the inductor 124 storing magnetic energy, and the amount of magnetic energy which can be stored by the inductor 124 is dependent on its inductance  $L_1$ , as will be well understood.

[0109] FIG. 7B shows a simplified representation of the voltage across the current sense resistor 615 against time  $t$ , again from the time  $t_0$  when the FET 608 is turned on. Shortly after the FET 608 is turned on, a voltage develops across the inductor 124, this being the back EMF generated by the inductor 124 as the inductor opposes the increase in current. At this time, therefore, the voltage across the current sense resistor 615 as shown in FIG. 7B is small, since almost all of the voltage difference provided by the DC supply 118 drops across the inductor 124. Then, as the current through the inductor 124 increases and the back EMF of the inductor 124 decays, the voltage across the current sense resistor 615 increases. This is seen as the development of a negative

voltage across the current sense resistor 615, as shown in FIG. 7B. That is, the voltage across the current sense resistor 615 becomes increasingly negative with the length of time that the FET 608 is on.

[0110] Since the increasingly negative voltage across the current sense resistor 615 corresponds with the increasing current through the inductor 124, the magnitude of the voltage across the current sense resistor 615 is indicative of the current flowing through the inductor 124. While the FET 608 remains on, the current through the inductor 124 and the voltage across the current sense resistor 615 increase substantially linearly towards respective maximum values  $I_{max}$ ,  $V_{max}$  (which are dependent on the DC voltage supplied by DC supply 118 and the DC resistance of the circuit 600) with a time constant dependent on the inductance  $L_1$  and on the DC resistance of the circuit 600. It should be noted that as the current through the inductor 124 is varying after time  $t_0$  to some inductive heating of the susceptor 132 may occur while the DC current through the first inductor 124 builds up.

[0111] The circuit 600 is configured such that the amount of energy which is stored in the first inductor 124 in the time during which the FET 608 is switched on, is determined by the control arrangement and can be controlled by the controller 1001. That is, the controller 1001 controls an amount of DC current (and thus an amount of magnetic energy) allowed to build up in the inductor 124, as will now be described.

[0112] As described above, the control voltage 1031 is applied to the control voltage line 619. In this example, the control voltage 1031 is a positive voltage and the voltage input to the positive terminal of the comparator 618 (i.e. the voltage at the fifth node 600E) at any one time is dependent on the value of control voltage 1031 and the voltage at the fourth node 600D. When the negative voltage across the current sense resistor 615 reaches a particular value, it cancels, at the fifth node 600E, the positive control voltage 1031 and gives a voltage of 0V (i.e. ground voltage) at the fifth node 600E. In this example, the resistor 617a has a resistance of 2 k $\Omega$ . The resistor 617b represents an effective resistance to the controller 1001 of 70 k $\Omega$ . The voltage at the fifth node 600E reaches 0V when the negative voltage across the current sense resistor 615 has the same magnitude as the control voltage 1031.

[0113] The comparator 618 is configured to compare the voltage at its positive terminal to the voltage of ground 616, connected to its negative terminal, and output a signal as a result. In one example the comparator is a standard component FAN 156, as may be obtained from On-Semiconductor. Accordingly, when the voltage at fifth node 600E reaches 0V, the comparator 618 receives a 0V signal at its positive terminal, and the result of the comparison by the comparator 618 is that the voltage at the positive terminal is equal to the voltage at the negative terminal. The comparator 618 consequently outputs a signal to the flip-flop 622 and causes the FET 608 to be switched off. As such, switching off of the FET 608 is dependent on a voltage condition detected in the circuit 600. Namely, in this example, when the comparator 618 detects by comparison of the voltage across its terminals that a negative voltage across the current sense resistor 615 has reached the same magnitude as the control voltage 1031, which occurs at time  $t_1$ , the FET 608 is switched off. In FIG. 7A, the DC current flowing through the inductor 124 at time  $t_1$  when the FET 608 is switched off is labelled  $I_1$ .

[0114] When the FET 608 is turned off, at time  $t_1$ , the FET 608 switches from acting like a closed switch to acting like a diode 608a in the resonator section 601, and for the purposes of supply from the DC supply 118 effectively acting like an open switch. At time  $t_1$  the path of the DC current through the inductor 124 to ground 616 is interrupted by the FET 608. This triggers the current flowing in the first inductor 124 to drop off (this is not shown in FIG. 7A), and the inductor 124 opposes this change in current by generating an induced voltage. Accordingly, current begins oscillating back and forth between the inductor 124 and the capacitors 606, 608 at the resonant frequency of the first resonator section 601.

[0115] Similarly, the voltage across the inductor 124 and thereby between the first FET drain 608D and source 608S terminals begins to oscillate at the resonant frequency of the first resonator section 601. As the current through and voltage across the inductor 124 begin to oscillate, the susceptor 132 is inductively heated. Switching the FET 608 to the OFF state, therefore acts to release the magnetic energy stored in the inductor 124 at time  $t_1$  to heat the susceptor 132.

[0116] FIG. 8 shows a trace 800 of the voltage across the first FET 608, starting from the FET 608 being in the ON state from time  $t_0$  to  $t_1$ . Over the time illustrated in FIG. 8 the first FET 608 is turned off and on twice.

[0117] The voltage trace 800 comprises a first section 800a between times  $t_0$  and  $t_1$  when the first FET 608 is ON, and a second section 800b to 800d when the first FET 608 is switched off. At 800e the FET 608 is switched on again, and a third section 800f which is equivalent to the first section 800a begins while the first FET 608 remains on and the above-described process of building up of DC current through the inductor 124 repeats. FIG. 8 also shows a fourth section 800g when the first FET 608 is switched off again to allow oscillation of the voltage across the FET 608, and a fifth section 800h when the first FET 608 is subsequently switched on again.

[0118] The voltage across the first FET 608 is zero when the first FET 608 is on in sections 800a, 800f and 800h. When the first FET 608 is turned off as indicated by section 800b to 800d and also by section 800g, the first inductor 124 uses the energy stored in its magnetic field (which magnetic field was the result of the DC current built up when the first FET 608 was on) to induce a voltage that opposes a drop in the current flowing through the first inductor 124 as a result of the first FET 608 being off. The voltage induced in the first inductor 124 causes a corresponding variation in voltage across the first FET 608. During this variation in voltage, the first inductor 124 and the capacitors 606, 610 begin to resonate with each other with a sinusoidal waveform. The voltage shown by voltage trace 800 initially increases (see for example 800b) as the induced voltage in the first inductor 124 increases to oppose a drop in current due to the first FET 608 being off, reaches a peak (see for example 800c) and then, as the energy stored in the magnetic field of the first inductor 124 diminishes, decreases back to zero (see for example 800d).

[0119] The varying voltage 800b to 800d and 800g produces a corresponding varying current (not shown) and, since during the off time of the first FET 608, the capacitors 606, 610 and the first inductor 124 act as a resonant LC circuit, the total impedance of the combination of the first inductor 124 and capacitors 606, 610 is at a minimum during

this time. It will therefore be understood that the maximum magnitude of the varying current flowing through the first inductor 124 will be relatively large. This relatively large varying current accordingly causes a relatively large varying magnetic field in the first inductor 124 which causes the susceptor 132 to generate heat. The time period over which the voltage across the first FET 608 varies as indicated by section 800b to 800d and by section 800g in this example depends on the resonant frequency of the first resonator section 601.

[0120] Referring now to FIG. 6 and FIG. 8, the circuit 600 is configured such that when the first FET 608 is off and the voltage across the first FET 608 decreases back towards 0V, the zero-voltage detector 621 detects this voltage condition and outputs a signal to the flip-flop 622 which causes the first FET 608 to be switched back to the ON state. That is, in response to this voltage condition detected within the first resonator section 601, the FET 608 is switched from the OFF state to the ON state. The zero-voltage detector 621 may be considered to detect a voltage condition indicative that a given proportion of a cycle of current oscillation between the inductive element and the capacitive element has been completed since the FET 608 was switched off. That is, the zero-voltage detector 621 detects that a half-cycle of current (and voltage) oscillation at the resonant frequency of the first resonator section 601 has been completed by the zero-voltage detector 621 detecting that the voltage across the FET 608 has returned to 0V or nearly 0V.

[0121] In some examples, the zero-voltage detector 621 may detect when the voltage across the first FET 608 has returned to at or below a voltage level 801 and as such may output a signal to cause switching of the state of the FET 608 before the voltage across the FET 608 reaches exactly 0V. As is illustrated by FIG. 8, the operation of the zero-voltage detector 621 curtails oscillations of the voltage in the resonator section 601 after one half-cycle and thus results in a substantially half-sine wave voltage profile across the first FET 608. Further details of the operation of the zero-voltage detector 621 will be described below with reference to FIG. 9.

[0122] When the first FET 608 is switched back on, at point 800e, a DC current driven by the DC source 118 again builds up through the first inductor 124. The first inductor 124 may then again store energy in the form of a magnetic field to be released when the first FET 608 is next switched off to initiate resonance within the first resonator section 601. As the first FET 608 is repeatedly switched on and off in this way, the above described process is continuously repeated to heat the susceptor 132.

[0123] It should be noted that the above described building up of current through the inductor 124 described with reference to FIGS. 7A and 7B occurs both when the FET 608 is turned on initially in response to a START signal 1011 from the controller 1001 and when the FET 608 is switched on subsequently by a zero-voltage condition detected by the zero-voltage detector 621. In the first instance, in response to the START signal 1011, the current in the inductor 124 builds up substantially linearly from 0. In the second instance, when the FET 608 is turned back on in response to a detected zero voltage condition at point 800e, some excess current is circulating in the circuit 600 (e.g. from previous cycles of switching on and off of the FET 608). As the FET 608 is turned back on following the detection of a zero-voltage condition, the recirculating current produces an

initial negative current through the FET 608. Then, while the FET 608 remains on, the current through the FET 608 and inductor 124 builds up, substantially linearly, from the initial negative current value produced by the recirculating current. As the current through the inductor 124 builds up, the voltage across the current sense resistor 615 correspondingly becomes increasingly negative, in the manner described above.

[0124] In examples, switching on and off of the FET 608 may occur at a frequency of around 100 kHz to 2 MHz, or around 500 kHz to 1 MHz, or around 300 kHz. The frequency at which the switching on and off of the FET 608 occurs is dependent upon the inductance L, the capacitance C, the DC supply voltage supplied by the supply 618 and further upon a degree to which current continues recirculating through the resonator section 601 and the loading effect of the susceptor 132. For example, where the DC supply voltage equals 3.6V, the inductance of the inductor 124 is 140 nH, and the capacitance of the resonator section 601 is 100 nF, the time for which the FET 608 remains on may be around 2700 ns and the time for a half-cycle of oscillation to complete when the FET 608 is off may be around 675 ns. These values correspond to a power of around 20 W being supplied from the DC voltage supply 118 to the resonator section 601. The above value of the time for which the FET 608 remains on is affected by the amount of current which recirculates in the circuit, since as described above, this recirculating current causes an initial negative current through the inductor upon switching on of the FET 608. It should also be noted that the time for the current to build up to the value which causes switching off of the FET 608 is also at least in part dependent on the resistance of the inductor 124, however, this has a relatively minor effect on the time when compared to the effect of the inductance of the resonator section 601. The time for a half-cycle of oscillation to complete (of in this example 675 ns) is dependent on the resonant frequency of the resonator section 601 which is affected not only by the values of inductance and capacitance of the inductor 124 and capacitors 606, 610 respectively, but also by the effective resistance provided by loading the inductor 124 with the susceptor 132.

[0125] Thus far, the circuit 600 has been described in terms of its operation to heat the susceptor 132 by one inductor, the first inductor 124, and thus only a part of the circuit 600 used by the device 100 has been described. However, as described above, the device 100 also comprises a second inductor 126 for heating the second zone 132b of the susceptor 132. FIG. 9 shows the circuit 600 comprising the second inductor 126 in addition to the first inductor 124.

[0126] As shown in FIG. 9, in addition to the features described with reference to FIGS. 6 to 8, the circuit 600 comprises a second resonator section 701 comprising the second inductor coil 126, a third capacitor 706, a fourth capacitor 710 and a second FET 708, having a drain terminal 708D, a source terminal 708S, and a gate terminal 708G. Additionally, the circuit 600 comprises a second gate driver 723 configured to provide a gate-source voltage to the second FET gate terminal 708G. The controller 1001 is not shown in FIG. 9 but the controller 1001 controls the circuit 600 in the manner described with reference to FIGS. 6 to 8 and additionally is configured to provide a control signal 1012 to the second gate driver 723. Some reference numer-

als of features of the circuit 600 already described with reference to FIG. 6 have been omitted from FIG. 9 for the sake of clarity.

[0127] As described above, the first inductor 124 is arranged to heat the first zone 132a of the susceptor 132 and the second inductor 126 is arranged to heat the second zone 132b of the susceptor 132. The second inductor 126, third and fourth capacitors 706, 710, and second FET 708 are arranged to form the second resonator section 701, in the same manner as the first inductor 124, first and second capacitors 606, 610, and first FET 608 are arranged to form the first resonator section 601. In one example, the third and fourth capacitors 706, 710 are also COG capacitors and may have a capacitance of around 100 nF. The second inductor 126 in one example has a DC resistance of around 8 mΩ. When active, the second resonator section 701 operates to heat the susceptor 132 in an equivalent manner as described above for the first resonator section 601 and description of this will not be repeated here.

[0128] It will be appreciated that the value of the DC resistance of the inductors 124, 126 will have an effect on the efficiency of the circuit 600, since a higher DC resistance will result in higher resistive losses in the inductor 124, 126 and as such it may be desirable to minimize inductor DC resistance, for example by changing the number of windings, or the cross-section of the inductors 124, 126. Furthermore, it will be appreciated that an AC resistance of the inductor 124 is increased compared to the DC resistance due to the skin effect. As such, the use of litz wire in examples provides for reducing the skin effect, and thereby reducing AC resistance and associated resistive losses from the inductors 124, 126. To give an example, where the first inductor 124 has a DC resistance of around 5 mΩ and the second inductor 126 has a DC resistance of around 8 mΩ, and the circuit operates at around 300 kHz, the particular arrangement of litz wire forming the coils results in effective resistances for the inductors 124, 126 of around 1.14 times their DC resistance values.

[0129] A node 700A in the second resonator section 701 is equivalent to the first node 600A of the first resonator section 601 and is electrically connected to the first node 600A and thereby to the positive terminal 118a of the DC supply 118. A node 700C is at the equivalent position in the second resonator section 701 as is the third node 600C of the first resonator section 601 and the node 700C is similarly connected to ground 616.

[0130] It is important to note that the circuit 600 is configured to be operated by the controller 1001 such that only one of the resonator sections 601, 701 is active at any one time. Examples of this operation will be described in more detail below.

[0131] During the activation of one of the resonator sections 601, 701, the zero-voltage detector 621 is configured to detect a zero-voltage condition in the active resonator section 601, 701 and thus control switching of the respective FET 608, 708 of the active resonator section 601, 701. The zero-voltage detector 621 controls when the respective FET 608, 708 of the active resonator section 601, 701, is switched back on (such as at point 800e), as will now be described in more detail, with reference to FIGS. 8 to 10.

[0132] In the circuit 600, the zero-voltage detector 621 is configured to detect a zero-voltage condition at the second node 600B of the first resonator section 601 or at the equivalent node 700B of the second resonator section 701.

When one of the first resonator section **601** and second resonator section **701** is active, the zero-voltage detector **621** detects each time the respective FET **608**, **708** has been switched off, that the voltage across that FET **608**, **708** has returned to zero (e.g. point **800e** in FIG. **8**) or, is close to zero e.g. below a level **801**. In response to the zero-voltage detector **621** making this detection, a signal is output to change the state of the flip-flop **622**. The respective gate driver **623** which is in operation then outputs a gate-source voltage to switch the respective FET back to the ON state.

[0133] A first small signal diode **725** connects the zero-voltage detector **621** to the first resonator section second node **600B** and a second small signal diode **726** connects the zero-voltage detector **621** to the equivalent node **700B** of the second resonator section **701**. Specifically, anodes of the first small signal diode **725** and second small signal diodes are connected to the zero-voltage detector **621** input via a common node **701B** while cathodes of the diodes **725**, **726** are connected respectively to the nodes **600B**, **700B**.

[0134] The operation of the zero-voltage detector **621**, in a particular example, will now be described with reference to FIG. **10**, which shows the zero-voltage detector **621** and the flip-flop **622**. In FIG. **10**, the components which make up the zero-voltage detector **621** are enclosed by a dotted line box. The node **701B** connected to the anodes of the first and second small signal diodes **725**, **726** is shown. The start signal **1011** from the controller **1001** to the zero-voltage detector **621** can also be seen in FIG. **13**.

[0135] The zero-voltage detector **621** in this example comprises an inverter gate **U103** having an input **2** from the node **701B** and an output **4** connected to an input of the flip-flop **622**. The inverter gate **U103** is powered by connections **5** and **3** and a capacitor **C108** isolates the connection **5** from ground. A logic power supply **621a** of, in this example, 2.5V is applied to the input **5** and via a pull-up resistor **R111** to the input **2** of the inverter gate **U103**. The logic power supply **621a** is in this example supplied from the controller **1001**. The inverter gate **U103** is configured to act as an OR gate for the START signal **1011** and a zero-voltage detection signal from the node **701B**. That is, the inverter gate **U103** is configured to receive a logic low signal in the form of the START signal **1011** from the controller **1001** to initiate operation of the circuit **600a**. The START signal **1011** may be provided by an “open drain” signal pin of the controller **1001**. The inverter gate **U103** is also configured to receive a logic low signal from the node **701B** when one of the first and second signal diodes **725**, **726** is forward biased due to one of the nodes **600B**, **700B** being at or near zero volts, as will be explained below. When either or both of these logic low signals is received by the inverter gate input **2** the inverter gate **U103** inverts the received signal and outputs a logic-high signal to the flip-flop **622**.

[0136] When the first inductor **124** is being operated to heat the susceptor **132**, the second FET **708** remains off. When the second FET **708** remains off, the second small signal diode **726** has either no bias or is reverse biased depending on the voltages at the logic power source and the DC supply **118**, that is, the voltage at a cathode end (nearest the node **700B**) of the second small signal diode **726** is either substantially the same as or higher than the voltage at an anode end (nearest the zero-voltage detector **621**) of the second small signal diode **726**.

[0137] During operation of the first resonator section **601**, when the first FET **608** is off and the voltage across it varies

as indicated by **800b-d** of FIG. **8**, the first small signal diode **725** is reverse biased. At the end of this variation in voltage, when the voltage reaches 0V as indicated by **800e**, or is close to 0V (e.g. at or below level **801**), the first small signal diode **725** becomes forward biased. Accordingly, when the first small signal diode **725** is forward biased at **800e**, the signal provided to the input **2** of the inverter gate **U103** becomes a logic low signal since a voltage drop is produced from the logic signal **621a** across the resistor **R111**. As such, once this logic low signal is inverted by the inverter gate **U103**, a logic high signal is provided at the output **4** of the inverter gate **U103**.

[0138] Although in the above description the functioning of the zero-voltage detector **621** is described in relation to controlling switching of the first FET **608**, it will be understood that the zero-voltage detector **621** functions in the same way, using the second small signal diode **726** instead of the first small signal diode **725**, to control the second FET **708**.

[0139] Still with reference to FIG. **10**, the flip-flop **622** comprises a clock input CLK, a reset input/RST, and an output Q. The flip-flop **622** also comprises further inputs D and VCC for supplying power, in this example the flip-flop receives the same 2.5V logic power supply **621a** from the controller **1001** as the inverter gate **U103** receives. The clock input CLK is connected to the output **4** of the inverter gate **U103** and is configured to receive a signal therefrom. When the output **4** of the inverter gate **U103** switches from logic-low to logic-high (due to the input **2** of inverter gate **U103** receiving a detected zero-voltage condition or a START signal **1011** as described above) the clock input CLK of the flip-flop **622** receives a logic-high rising edge signal which “clocks” the flip-flop **622** and makes the state of the flip-flop output Q high. The flip-flop **622** comprises a further input/RST configured to receive a signal from the output of the comparator **618**, by which the comparator **618** can change the state of the flip-flop **621** to cause the flip-flop output Q to be low. The flip-flop output Q is connected to the first and second gate drivers **623**, **723** and on receiving a high output from the flip-flop output Q, whichever one of the gate drivers **623**, **723** is active (due to receiving a signal **1021**, **1022** as described above) provides a gate driver signal to its respective FET **608**, **708**.

[0140] In one particular example, the flip-flop **622** may switch at half of the voltage of the logic power source **621a**, that is, at 1.25V in this example. This means that the forward bias voltage of the first small signal diode **725** and the voltage at the first FET drain **608D** must sum to 1.25V in order that the first FET **608** is switched on. In this example therefore, the first FET **608** is switched on when its drain **608D** is at 0.55V rather than at exactly 0V. It should be noted that ideally, switching may occur at 0V across the FET **608** for maximum efficiency. This zero-voltage switching advantageously prevents the first FET **608** from discharging the capacitors **606**, **610** and thereby wasting energy stored in said capacitors **606**, **610**.

[0141] FIG. **11** shows in more detail the first and second gate drivers **623**, **723** and their connection to the gates **608G**, **708G** of their respective FETs **608**, **708**. Each of the gate drivers **623**, **723** has an input IN which is configured to receive a signal dependent on the heater activation signals **1021**, **1022** supplied from the controller **1001**. Additionally, the signals received by the inputs IN of the gate drivers **623**, **723** are dependent on whether the signal output by the



flip-flop output Q is high. The inputs IN are connected to the output Q of the flip-flop 622 via respective resistors R125, R128, which in this example each have a value of 2 k $\Omega$ .

[0142] The gate drivers 623, 723 each have two further inputs VDD and XREF wherein each input VDD receives a 6V power supply from the controller 1001 and XREF receives a 2.5V logic voltage, which in this example is the same logic voltage supplied by the controller 1001 to the flip-flop 622 and inverter gate U103. The inputs VDD of each of the first and second gate drivers 623, 723 are connected to a 6V supply voltage and the inputs VDD are connected to ground via two buffering capacitors C120, C121. The gate drivers 623, 723 also each have a terminal GND connected to ground wherein the terminals VDD and GND act to supply power to the gate drivers 623, 723. In this example, the capacitors C120, C121 each have a value of 1  $\mu$ F. The gate drivers 623, 723 are configured to output gate drive voltages from respective outputs OUT. The outputs OUT of the gate drivers 623, 723 are connected respectively to FET gates 608G, 708G via resistors R114, R115, which in this example each have a resistance of 4.992.

[0143] Each gate driver 623, 723 is configured to receive a signal at its input IN to cause the gate driver to be activated only while a logic-high signal is provided from the flip-flop output Q and a heater activation signal 1021, 1022 is received from the controller 1001. An “open-drain” signal pin may be provided on the controller 1001 which is configured to provide the signals 1021, 1022.

[0144] In examples, initiation of the circuit 600 for heating by one of the resonator sections 601, 701 proceeds by the controller 1001 first initiating the desired one of the gate drivers 623, 723 by a respective one of the heater initiation signals 1021, 1022. The controller 1001 then supplies the START signal 1011 to the zero-voltage detector 621. The duration of the START signal 1011 should be shorter than the period of half a cycle of oscillation by the active resonator section 601, 701 (this period may be referred to as the “resonant fly-back period”). This allows the circuit to properly begin self-oscillating in response to a detected zero-voltage condition. In another example, the order the START signal 1011 and respective heater enable signal 1021, 1022 may be reversed such that the START signal 1011 is first applied to set the flip-flop Q output to high, and one of the heater initiation signal 1021, 1022 then applied to begin the self-oscillating of the resonator section 601, 701 corresponding to heater to which the signal 1021, 1022 is supplied.

[0145] To continue with describing in more detail the control arrangement for controlling the circuit 600, FIG. 12 shows a portion of the control arrangement comprising the comparator 618 and associated components. In FIG. 12, the positive terminal 118a of the DC power supply 118 is shown connecting to a node 1500A which is connected to nodes 600A, 700A of the first and second resonator sections 601, 701 respectively. The negative terminal 118b of the DC power supply is connected to a node 1500B which is equivalent to the node 600D shown in FIG. 6. The node 1500B connects to ground 616 via the current sense resistor 615. Between the nodes 1500A and 1500B an arrangement of capacitors C111, C112, C115 and C116, each in this example having a capacitance of 100  $\mu$ F, are connected in parallel, providing buffering between nodes 1500A and 1500B.

[0146] FIG. 12 shows in more detail components associated with the functioning of the comparator 618 for detecting that the current through the active inductor 124 or 126 has reached a given level. As described with reference to earlier figures, the comparator 618 acts to compare a voltage indicative of an amount of DC current flowing in the active inductor (124 or 126) to a control voltage 1031 originating from the controller 1001. The comparator 618 receives power via an input 6 which is connected via a 100 $\Omega$  resistor R116 to a 2.5V logic power signal, in this example supplied by the controller 1001 and the same logic signal as the signal 621a received by the flip-flop 622. The comparator power input 6 is connected to ground via a 10 nF capacitor C119. A further terminal 2 of the comparator 618 is connected directly to ground.

[0147] In some examples, the controller 1001 is a micro-processing unit comprising a timer (not shown) for generating a signal which produces the control voltage 1031. In this example, the control voltage 1031 is produced by a pulse-width modulated signal PWM\_DAC generated by the controller 1001. The timer of the controller 1001 generates a pulse-width modulated square waveform, with, for example, a magnitude of around 2.5V and a frequency of around 20 kHz and having a particular duty cycle. The pulse-width modulated signal PWM\_DAC is filtered by 10 nF capacitors C127 and C128, and by two 49.9 k $\Omega$  resistors R121, R123 and a 10 k $\Omega$  resistor R124 to provide a substantially constant control voltage 1031 at the frequency at which the controller 1001 controls the control voltage 1031 (of, e.g., around 64 Hz in examples). To adjust the control voltage 1031, the controller 1001 in examples is configured to adjust the duty cycle of the pulse-width modulated signal PWM\_DAC applied to the circuit 600. As such, the components positioned between the input PWM\_DAC and the positive terminal of the comparator 618 effectively provide for the control voltage 1031 to be produced by a pulse-wave modulated signal and for the control voltage 1031 magnitude to be adjusted by adjusting the duty cycle of this pulse wave modulated signal. The control voltage line 619 shown in FIGS. 6 and 9 may thus be replaced by these components. However, in other examples, the control voltage 1031 may be produced by a substantially constant voltage supplied, for example, by the controller 1001. In such examples, some or all of the components for shown in FIG. 12 for filtering the signal PWM\_DAC may not be present.

[0148] The node 1500B which is input to the comparator 618 positive input is, as mentioned above, equivalent to the node 600D of the circuit 600. It can be seen from FIG. 12 that, as described with reference to FIG. 6, the node 1500B is connected via the resistor 617a to the positive input of the comparator 618. As such, the operation of the comparator 618 is as described above: to receive an input at its positive terminal which is dependent on the control voltage 1031 and the voltage across the current sense resistor 615. When the voltage at the positive terminal of the comparator 618 reaches ground voltage, a signal /FF RST is output, via a resistor R118, to the flip-flop 622 input/RST to change the state of the flip-flop 622 and thereby switch the active FET 608/708 off.

[0149] FIG. 13 shows further components for a particular example of the control arrangement for the circuit 600. The components shown in FIG. 13 define current sense apparatus 1300 for providing a signal I\_SENSE indicative of an

amount of current drawn from the DC voltage supply **118** during operation of the circuit **600**. From this signal, the controller **1001** may determine a current drawn from the voltage supply **118**, and may use this along with a value of the voltage supplied by the DC voltage supply **118** to determine a value for a power supplied to the circuit **600**. In some examples, as will be described below, a determined value of power may be used by the controller **1001** for controlling the circuit **600**.

**[0150]** An input **1301** to the current sense apparatus **1300** is provided via the resistor **R120** shown in FIG. **12**. The input is therefore connected to the node **1500B** via the resistor **R120** and receives a voltage indicative of the voltage across the current sense resistor **615**. The current sense apparatus **1300** operates as a low-side current sensing apparatus for the circuit **600**. In that regard, the current sense apparatus **1300** comprises an op-amp **U110** running on a voltage of 3.8V supplied to an input **5** of the op-amp **U110** (component type TS507) set up for low-side current sensing using the current sense resistor **615**, as will be well understood. A transistor **U109** with built-in bias resistor (component type RN4986) acts to switch a 2.5V supplied by the controller **1001** up to the 3.8V supply for the op-amp **U110**. The power supply line from the transistor component **U109** is connected to ground via a 10 nF capacitor **C132**. Further, a 1 k $\Omega$  resistor **R130** is connected between the positive input of the op-amp **U110** and ground and a 412 k $\Omega$  resistor **R129** is connected between the 2.5V input from the controller **1001** and the positive input of the comparator **U110**. The negative terminal of the op-amp **U110** receives a voltage dependent on the voltage across the current sense resistor **615**. A resistor **R131** and capacitor **C133** in series provide filtering of the voltage signal received via the input **1301**. A further resistor **R133** (in this example having resistance of 97.6 k $\Omega$ ) and a 10 nF capacitor **C134** are connected in parallel between the input to the negative terminal of the op-amp **U110** and the output of the op-amp **U110** such that op-amp operates in a closed-loop mode.

**[0151]** The position of the current sense resistor **615**, which, as mentioned above, in one example is a 2 m $\Omega$  resistor, in the circuit allows for a plurality of parameters to be measured with one current sense resistor, which may allow for good efficiency. That is, the position of the current sense resistor **615** in the circuit allows measurement of: the FET peak current, which may be used, for example, in control of the induction heating power of the circuit; the average current out of the battery, which may be used in monitoring discharge of the battery and setting the induction power; and the average current into the battery, which may be used, for example, in monitoring charging of the battery.

**[0152]** The op-amp **U110** operates to output a voltage signal **I\_SENSE** to the controller **1001** which is indicative of the current through the current sense resistor **615** and thus allows the controller **1001** to determine the current drawn from the DC voltage supply **118** through the circuit **600**.

**[0153]** It should be noted that having regard to the first and second FETs **608** and **708**, and the topology of the circuit **600**, the phasing of the first and second inductor coils **124** and **126** with respect to each other may be chosen such that when the first inductor coil **124** is being operated, current sufficient to cause significant heating of the susceptor **132** is prevented from flowing in the second inductor coil **126**, and when the second inductor coil **126** is being operated, current

sufficient to cause significant heating of the susceptor **132** is prevented from flowing in the first inductor coil **124**.

**[0154]** As described above, the first **608** and second **708** FETs effectively act as diodes **608a**, **708a** when switched off and so may conduct a current if they are forward biased (i.e. the FETs are not perfect switches). Accordingly, in examples the circuit **600** may be configured so that when one of the first **124** and **126** inductor coils is active to heat the susceptor **132**, the voltage induced across the other inactive inductor coil does not forward bias the intrinsic diode of the FET associated with that inactive inductor coil but instead reverse biases it.

**[0155]** The effect of the above described control arrangement being configured to control the switching arrangements **608**, **708** of the circuit **600** in response to detected voltage conditions is that when one of the first resonator section **601** and the second resonator section **701** is active (i.e. its gate driver **623**, **723** is activated by the controller **1001**) that resonator section “self-oscillates”, while the other section remains inactive. That is, switching of the respective FET **608**, **708** in the resonator section **601**, **701** repeats at a high frequency as a first voltage condition (detected by the comparator **618**) causes the FET to be switched from on to off, and a second voltage condition (detected by the zero-voltage detector **621**) causes the FET to be switched from off to on.

**[0156]** The controller **1001** is configured to control the induction heating circuit **600** of the device **100** such that only one of the first inductor **124** and the second inductor **126** is active at any one time. The controller **1001** is configured to determine at a predetermined frequency which of the first inductor **124** and the second inductor **126** to activate.

**[0157]** In examples, during usage of the device **100** the controller **1001** determines at the predetermined frequency, i.e. one time for each of a plurality of predetermined time intervals, which of the first resonator section **601** and the second resonator section **701** to activate. In one example, each time the controller **1001** determines which of the first resonator section **601** and the second resonator section **701** to activate, the controller **1001** may determine to activate that resonator section to heat the susceptor **132** for the duration of the next predetermined interval. That is, where the predetermined frequency (which may be referred to as an “interrupt rate”) is 64 Hz, for example, the controller **1001** may determine at predetermined intervals of  $\frac{1}{64}$ s, which resonator section **601**, **701** to activate for a following duration of  $\frac{1}{64}$ s until the controller makes the next determination of which resonator section **601**, **701**, at the end of the following  $\frac{1}{64}$ s interval. In other examples, the interrupt rate may be, for example, from 20 Hz to 80 Hz, or correspondingly the predetermined intervals may be of length  $\frac{1}{80}$ s to  $\frac{1}{20}$ s. In order to determine which inductor **124**, **126** is to be activated for a predetermined interval, the controller **1001** determines which susceptor zone **132a**, **132b** should be heated for that predetermined interval. In examples, the controller **1001** determines which susceptor zone **132a**, **132b** should be heated with reference to a measured temperature of the susceptor zones **132a**, **132b**, as will be described below.

**[0158]** FIG. **14** shows a flowchart representation of an example method of determining which of the two resonator sections **601**, **701**, should be activated for a particular predetermined interval. In this example, the controller **1001**,

determines which of the first **601** and second **701** resonator sections to activate for the predetermined interval based on a present temperature **T1** of the first susceptor zone **132a** heated by the first inductor **124** and a present temperature **T2** of a second susceptor zone **132b** heated by the second inductor **126**. In an example, the present temperatures **T1** and **T2** of the first **132a** and second **132b** susceptor zones may be measured by respective thermocouples (examples of which are described below) attached to each zone of the susceptor **132**. The thermocouples provide an input to the controller **1001** allowing the controller **1001** to determine the temperatures **T1**, **T2**. In other examples, other suitable means may be used to determine the respective temperatures of the susceptor zones **132a**, **132b**.

[**0159**] At block **1051**, the controller **1001** determines a present value of the temperature **T1** and compares this to a target temperature **target1** for the first zone **132a** arranged to be heated by the first inductor **124**. The target temperature **target1** of the first zone **132a** has a value which may vary throughout a usage session of the device employing the circuit **600**. For example, a temperature profile may be defined for the first zone defining values for **target1** throughout a usage session of the device **100**.

[**0160**] At block **1052** the controller **1001** performs the same operation as was performed for the first inductor **124** at block **1051** and determines whether the present temperature **T2** of the second zone **132b** is below the target temperature **target2** for the second zone **132b** at this time. Again, the target temperature of the second zone **132b** may be defined by a temperature profile defining values of **target2** throughout a usage session. The temperature of the second zone **132b** may, similarly to the first zone **132a**, be measured by any suitable means such as by a thermocouple.

[**0161**] If the answers at block **1051** and block **1052** are both “no”, i.e. both susceptor zones **132a**, **132b** are presently at or above their respective target temperatures **target1**, **target2**, then the controller **1001** determines neither of the first and second resonator sections **601**, **701** should be activated for the next predetermined interval.

[**0162**] If the answer at block **1051** is “no” and the answer at block **1052** is “yes”, i.e. the first zone **132a** is at or above its target temperature **target1** but the second zone **132b** is below its target temperature **target2**, then the controller **1001** determines that the second resonator section **701** should be activated to heat the second zone **132b** for the next predetermined interval.

[**0163**] If the answer at block **1051** is “yes” and the answer at block **1052** is “no”, i.e. the first zone **132a** is below its target temperature **target1** and the second zone **132b** is at or above its target temperature **target2**, then the controller **1001** determines that the first resonator section **601** should be activated to heat the first zone **132a** for the next predetermined interval.

[**0164**] If the answer at block **1051** is “yes” and the answer at block **1052** is “yes”, i.e. both the first **132a** and second **132b** zones are below their respective target temperatures **target1**, **target2**, then the controller **1001** continues to block **1053**. At block **1053** the controller **1001** effectively acts to alternate activation of the first resonator section **601** and second resonator section **701** for each predetermined interval that both zones **132a**, **132b** remain below their respective target temperatures.

[**0165**] In some examples, in order to alternately activate the first **601** and second **701** resonator sections, at block

**1053** the controller **1001** determines if an even number of predetermined intervals has elapsed since the start of the session. If an even number of predetermined intervals has elapsed since the start of the session then the controller **1001** determines that the first resonator section **601** should be activated for the next interval. If an odd number of predetermined intervals has elapsed since the start of the session then the controller **1001** determines that the second resonator section **701** should be activated for the next interval. In other examples, it should be understood, that the controller **1001** may instead activate the second resonator section **701** when an even number of intervals has elapsed and the first resonator section **601** when an odd number of intervals has elapsed.

[**0166**] In certain examples, the circuit **600** is configured such that once one of the resonator sections **601**, **701** is activated by receipt of a signal **1021** or **1022** at one of the gate drivers **623**, **624**, the that resonator section **601/701** continues to operate, i.e. self-oscillate, until deactivated by the controller **1001**, for example by providing a different signal to the gate driver of that resonator section **601/701**. As such, upon determining which of the resonator sections **601**, **701** to activate during a given interval, the controller **1001**, in order to initiate this activation may deactivate one of the resonator sections **601**, **701** which was active during a previous interval.

[**0167**] To illustrate an example of block **1053** where method **1050** shown in FIG. **14** is performed with intervals of  $\frac{1}{64}$ s, if the controller **1001** determines that both zones **132a**, **132b** are below their respective target temperatures **target1**, **target2** and an even number of  $\frac{1}{64}$ s intervals has elapsed since the start of the usage session of the device **100**, then the controller **1001** activates the first resonator section **601** for the next  $\frac{1}{64}$ s interval while the second resonator section **701** is rendered inactive, which in examples requires the controller **1001** deactivating the second resonator section **701**. If after this next interval of  $\frac{1}{64}$ s both zones **132a**, **132b** remain below their respective target temperatures **target1**, **target2**, then for the following  $\frac{1}{64}$ s interval the controller **1001** activates the second resonator section **701** while the first resonator section **601** is rendered inactive, which in examples requires the controller **1001** deactivating the second resonator section **701**. For each interval in which both zones **132a**, **132b** remain below their respective target temperatures this alternating between activating the first **601** and second **701** resonator sections continues.

[**0168**] Altogether, the method **1050** has the effect that the two inductors **124**, **126** are never activated at the same time. Where it is determined that both inductors **124**, **126** require activation to bring their respective zones **132a**, **132b** to target temperature the controller **1001** alternates the supply of power to the inductors **124**, **126** at the predetermined frequency to bring both zones **132a**, **132b** up to their respective target temperature. Therefore, during a particular point in a usage session, activation of the inductor coils **124**, **126** to heat their respective susceptor zone **132a**, **132b** may be alternating at a particular frequency, such as 64 Hz. It can be seen that, for example, during a period of a usage session comprising a plurality of intervals where the first zone **132a**, is substantially below its target temperature and the second zone **132b** is at or above its target temperature, the method **1050** has the effect that power may be supplied to the first resonator section **601** for close to 100% of this period. However, for a period of a usage session comprising

a plurality of intervals in which both zones **132a**, **132b** are below their target temperatures, each inductor may receive power for roughly 50% of this period.

[0169] In examples, the controller **1001** is also configured at predetermined intervals that coincide with the predetermined intervals at which the method **1050** is performed, to determine a power being supplied to one of the resonator sections **601**, **701** from the DC supply **118**.

[0170] As described above, with reference to FIGS. **9** to **11** in particular, in order to control which of the first resonator section **601** and second resonator section **701** is active at any one time, the controller **1001** as well as transmitting a START signal **1001** to initiate operation of the circuit **600** is configured to selectively transmit a first heater operation signal **1011** to the first gate driver **623** to activate the first resonator section **601** or a second heater operation signal **1012** to the second gate driver **723** to activate the second resonator section **701**.

[0171] For example, when the controller **1001** initiates operation of the circuit **600** and the controller **1001** transmits the first heater operation signal **1011**, the circuit **600** operates as described above to activate the first inductor **124** to heat the first susceptor zone **132a**. When the controller **1001** transmits the second heater operation signal **1012** the circuit **600** operates to activate the second inductor **126** to heat the second susceptor zone **132b**. If the controller **1001** transmits neither of the first heater signal **1011** and the second heater signal **1012** then neither inductor **124**, **126** is activated and the susceptor **132** is not heated.

[0172] The controller **1001** is configured to control the power supplied from the DC voltage supply **118** to the circuit **600** for inductive heating of the susceptor **132** based on a comparison of a measurement of power supplied to the circuit **600** and a target power. The controller **1001** is configured to control the power supplied to the circuit **600** by controlling the switching arrangement of the circuit **600**, i.e. by controlling switching of the FETs **608**, **708**. The controller **1001** may control switching of the FETs **608**, **708** by setting the control voltage **1031** which determines the DC current which is allowed to build up in the inductor **124**, **126** corresponding to that FET **608**, **708** before the FET **608**, **708** is switched off.

[0173] FIG. **15** shows an example method **1100** performed by the controller **1001** to control the power supplied to the circuit **600**. At block **1101** the controller **1001** determines the power **P** supplied from the DC supply **118** to the circuit **600**. For example the controller **1001** may determine an average of the power supplied to the circuit **600** during the previous predetermined interval. In examples, the power **P** being supplied to the circuit **600** during an interval may be determined by measurement of the voltage across and DC current being driven through a given one of the resonator sections **601**, **701**. The controller **1001** may then determine the product of the voltage across and DC current through the given one of the resonator sections **601**, **701** to determine the power **P** supplied to that resonator section **601**, **701**.

[0174] In examples, the determined power **P** is an average power supplied from the DC supply **118** over the predetermined interval which may be determined by determining a product of the average DC voltage across the DC supply **118** and the average DC current drawn from the DC supply **118** over the previous interval.

[0175] In the example device **100**, the DC supply **118** is a battery which is connected to the controller **1001**, and the

controller **1001** then outputs the voltage of the DC supply **118** to the circuit **600**. The controller **1001** is configured to determine the DC voltage supplied by the battery **118**. The current drawn from the battery **118** is determined by the operation of the current sense apparatus **1300**. The controller **1001** determines the DC voltage and DC current once for every  $\frac{1}{64s}$  interval. The DC voltage can be considered to be essentially constant over this short time period. However, the current is varying at a rate dependent on the rapid rate of switching on and off the circuit. As described above, this is around 300 kHz in some examples. The current sense apparatus **1300** as described above with reference to FIG. **13**, outputs a signal **I\_SENSE** which is filtered to remove this around 300 kHz signal. An average DC current for the  $\frac{1}{64s}$  interval is therefore obtained by taking a measurement of this filtered signal **I\_SENSE**, and the measurement of **I\_SENSE** is taken just before the end of the  $\frac{1}{64s}$  interval in order to allow the signal from the filter to settle. The controller **1001** thereby obtains a DC voltage and DC current measurement for the  $\frac{1}{64s}$  interval and can calculate a product of these values to obtain the determined power **P**. This determined power **P** may be considered to be an average of the power supplied by the DC supply **118** over the  $\frac{1}{64s}$  interval.

[0176] At block **1102** the supplied power **P** determined at block **1101** is compared to a target power. Where the determined power **P** is an average power over the predetermined interval, the target power is a target average power over the same interval. In one example, the target power is a target for the average power supplied over the predetermined interval and may have a value of between 10 and 25 W, or between 15 and 23 W, or around 20 W. In this example, the target power is a range, for example, 20-21 W or of 15-25 W. The controller **1001** may accordingly at block **1102** compare the supplied power value **P** determined at block **1101** to the target range and determine whether the supplied power is below the range, within the target range, or above the target range. For example, where the target range is 20-21 W, at block **1102** the controller **1001** determines whether  $P < 20$  W, or  $20 W \leq P \leq 21$  W, or  $P > 21$  W.

[0177] Based on the comparison of the supplied power **P** to the target range, the controller **1001** determines whether and how to adjust the power for the next predetermined interval with the aim of bringing the actual power supplied to the active inductor **124** or **126** during the next predetermined interval towards the target power range. That is, if the supplied power **P** is below the target range, then the controller **1001** determines to increase the power supplied to the circuit **600** over the next predetermined interval. If the supplied power **P** is above the target range, then the controller **1001** determines to decrease the power supplied to the circuit **600** over the next predetermined interval. If the supplied power **P** is below the target range, then the controller **1001** determines not to adjust the power supplied to the circuit **600** over the next predetermined interval.

[0178] Due to the configuration of the circuit **600** described above, the supplied power **P** for a given predetermined interval is dependent on the value of the control voltage **1031** for that interval. Taking the example of one  $\frac{1}{64s}$  interval for which the first resonator section **601** is active, this  $\frac{1}{64s}$  interval comprises many repeating cycles comprising sections **800a** to **800e** of the voltage trace **800** and repeats thereof. For each cycle during the period of time  $t_1$  to  $t_0$  the resonator section **601** is allowed to resonate, and

since for this period the FET 608 is off, no power is drawn from the DC supply 118 through the first resonator section 601. Substantially all of the power drawn from the DC supply 118 during the given  $\frac{1}{64}$ s interval to power the resonator section 601 is thus drawn during the period between  $t_0$  and  $t_1$  while the inductor 124 is being “energized” with current, i.e. while the FET 608 is on. The time between  $t_1$  and to is determined by the resonant frequency of the first resonator section 601. This resonant frequency may remain substantially constant, at least throughout a given  $\frac{1}{64}$ s interval (although may vary over the period of operation of the circuit 600 due to dependence on coil and suscepter temperature and battery voltage). The length of time  $t_0$  to  $t_1$  is determined by the value of the control voltage 1031, as well as the DC voltage supplied by the DC supply 118 and the resistance and inductance of the first resonator section 601 (the same applying for the second resonator section 701). That is, for a given DC supply voltage, the control voltage 1031 sets the current  $I_1$  which is allowed to build up in the inductor 124 between  $t_0$  and  $t_1$ , but where the DC supply voltage is reduced, the time required to build up a given value of  $I_1$  is increased. As such, the average power supplied during the  $\frac{1}{64}$ s interval is dependent on the value of the control voltage 1031.

[0179] In examples, therefore, in order to control the power supplied to the circuit 600 during the next interval the controller 1001 sets the value of the control voltage 1031 for the next interval. In examples, for a given DC supply 118 over a predetermined interval during which one of the resonator sections 601, 701 is active, a larger positive value of the control voltage 1031 results in a larger value of power P being delivered to the circuit 600. Therefore, in such examples, where the controller 1001 determines that the supplied power P over the last interval was above the target range, the controller 1001 reduces the control voltage 1031 for the next interval. Where the controller 1001 determines that the supplied power P over the last interval was below the target range, the controller 1001 increases the control voltage 1031 for the next interval. And, where the controller 1001 determines that the supplied power P over the last interval was above the target range, the controller 1001 leaves the control voltage 1031 unchanged for the next interval.

[0180] It should be noted that in one example of the above method 1100 the power supplied P is determined at block 1101 is a power supplied to a particular one of the resonator sections 601, 701. For example, the power P may be determined by measuring the voltage across the first resonator section 601 and the DC current through the first resonator section 601. In such an example, it is the power P supplied to the first resonator section 601 which is used to control the control voltage 1031. It should also be noted that for a given control voltage 1031, in some examples, the power supplied to each of the inductors 124, 126 when the respective resonator sections 601, 701 are active may be different. This may be, for example, because the inductors 124, 126 have different values of inductance or DC resistance, or the capacitance of the two resonator sections 601, 701 is not equal. Therefore, in this example, during a given predetermined interval, a target power outside of the target power range may be supplied to the second resonator section 701 but since the control voltage 1031 is controlled based on

the power P supplied to the first resonator section 601, in this example the controller 1001 may not adjust the control voltage 1031.

[0181] For example, for a given value of the control voltage 1031, the controller 1001 may determine at block 1101 that an average power of 20 W was supplied to the first resonator section 601 over a given interval, with the target voltage in this example being 20-21 W. At block 1102 the controller 1001 determines that the supplied voltage was within the target range and accordingly the controller 1001 determines not to adjust the control voltage 1031. Consider that for the next predetermined interval, the controller 1001 determines (by the example method 1050) that the second resonator section 701 and not the first resonator section 601 is to be activated. For the given value of the control voltage 1031, in this example, 22.5 W is delivered due to differences in the electrical properties of the first 601 and second 701 resonator sections. However, in this example, at block 1102 the controller 1001 compares the last measured value of power P delivered to the first resonator section 601 and therefore determines at block 1103 not to adjust the control voltage 1031. As such, in an example of the method 1100 the actual power supplied to the circuit 600 may be outside of the target range. However, this may allow for controlling the power supplied to the inductors 124, 126 by measuring only the power P supplied to one of the resonator sections 601, 701. This may provide a simple and useful solution to maintain the power supplied to the circuit 600 to within an acceptable range if, for example, the resonator sections 601, 701 and components thereof have roughly similar electrical properties.

[0182] As mentioned above, in some examples, the DC supply 118 is a battery with a voltage of around 2 to 10V, or 3 to 5V, or in one example around 4.2V. In some examples, the DC voltage produced by the DC supply 118 may change, e.g. decrease, over the time that the circuit 600 is operated. For example, where the DC voltage source 118 is a battery, the battery may initially supply a voltage of 4.2V but the voltage supplied by the battery may reduce as the battery depletes. After a given period, therefore, the DC voltage source 118 may supply, for example, 3.5V instead of an initial 4.2V.

[0183] As described above, at a given supply voltage, the value of the control voltage 1031 controls the amount of current which is allowed to build up in the active inductor 124/126 before the respective FET 608/708 is switched off. Power is supplied from the DC voltage supply 118 to “energize” the active inductor 124/126 by allowing a build-up of DC current when the FET 608, 708 is on. As was also described above, the time  $t_1$  for the current to build up to the value which causes switching of the FET 608/708 is dependent on the DC voltage supply. Therefore, for example, if the voltage supplied by the DC supply 118 reduces, the rate at which current builds up in the inductor coil 124 reduces, resulting in reduced power P being supplied to the circuit 600.

[0184] The example method 1100 may provide for a target power to be maintained even in the event that the supplied voltage from the DC supply 118 changes. That is, since an actual supplied power P is determined and used to control the control voltage 1031, the controller 1001 can act to maintain a target power by adjusting the control voltage 1031. For example, where the battery level has depleted, the controller 1001 measures that the power P supplied to the

circuit 600 at a given control voltage 1031 has reduced, and acts to increase the power P supplied to the circuit by increasing the control voltage 1031. As such, a target power level may be maintained while a battery used to power the circuit 600 depletes. This is advantageous since maintaining a target power level may provide for optimal efficiency of operation of the induction heating circuit 600. For example, maintaining a substantially constant power supplied allows for consistent heating of the aerosolizable material 110a regardless of supply voltage. Similarly, the example method 1100 provides for providing a substantially constant power regardless of other changing factors in the circuit which might affect the amount of power delivered, such as different loading on the circuit 600 being provided by the susceptor 132 when the susceptor 132 temperature increases. This provides a consistently good experience for the consumer, for example by providing a consistent time to first puff, i.e. a consistent time between the device 100 being activated and being ready to provide aerosol to be inhaled by the user.

[0185] In another example, the measured power value P upon which control of the control voltage 1031 is based is changed throughout a usage session. For example, during a particular usage session, for a first part of the usage session (e.g. a first ~60 s of the usage session), the temperature profile may be such that the first inductor 124 is primarily active, while the second inductor 126 is inactive. For this first part of the usage session it may be appropriate to base control of the control voltage 1031 on measurements of the power delivered to the first resonator section 601. However, later in the session, again e.g. due to the temperature profile for the session, it may be that the second inductor 126 is primarily active, while the first inductor 124 is active for less of the time. Thus, for a second part of the usage session (e.g. after ~60 s), it may be advantageous to control the control voltage 1031 based on the measurements of power delivered to the second resonator section 701. The controller 1001 may accordingly switch from basing control of the control voltage 1031 on measurements of power supplied to the first resonator section 601 to basing control of the control voltage 1031 on measurements of power supplied to the second resonator section 701. In this way, the target power may be more closely adhered to throughout a usage session, since, for example, the control voltage 1031 is being set based on a comparison of the actual power being delivered to the active inductor 124, 126 to the target power range.

[0186] In some examples, where the controller 1001 determines at block 1103 that the power should be adjusted, the controller 1001 may adjust the control voltage 1031 in predetermined steps. For example, the controller 1001 may be configured to adjust the control voltage 1031 by a predetermined amount per predetermined time interval. Where at block 1102 the controller 1001 determines that the supplied power P was below the target power range the controller 1001 may increase the control voltage 1031 by a predetermined number of volts for the next predetermined interval. Conversely, where at block 1102 the controller 1001 determines that the supplied power was above the target power range the controller 1001 may increase the control voltage 1031 by a predetermined amount for the next predetermined interval.

[0187] In the example described above with reference in particular to FIG. 12, the control voltage 1031 is produced by a pulse-wave modulated signal PWM\_DAC. The signal PWM\_DAC, as described above has a rectangular wave-

form at 2.5V. The duty signal of the signal PWM\_DAC is controllable by the controller 1001 which sets a value of 0 to 800 for the PWM\_DAC duty cycle, this value corresponding to a duty cycle of 0% at 0 and 100% at 800. The signal PWM\_DAC when filtered provides the substantially constant control voltage 1031 and therefore the settings of from 0 to 800 of the duty cycle of the PWM\_DAC signal provide for the control voltage 1031 to have a magnitude of from 0 to 2.5V. In this example, the controller 1001 may adjust the duty cycle setting of the PWM\_DAC signal by a set amount, such as 8 out of 800, or leave the setting unchanged, for each predetermined interval. In another example, the controller 1001 may provide for the control voltage 1031 to be adjusted by some other means, and if the controller 1001 determines that the control voltage 1031 should be adjusted, the controller 1001 may adjust the control voltage 1031 by, e.g., 1%, or 2%, or 5% of the maximum value of the control voltage 1031 for the next predetermined interval.

[0188] In some examples, when operation of the circuit 600 is initiated by the controller 1001, e.g. to start a use session of the device 100 comprising the circuit 600, the control voltage 1031 is set to a predetermined initial value. In one example, a value of the control voltage 1031 (for example, a duty cycle setting of the signal PWM\_DAC which produces this value of the control voltage 1031) which corresponds with a target power level is determined during setup of the circuit 600. That is, the power delivered to the circuit 600 may be determined (e.g. measured or determined theoretically) for a number of values of the control voltage 1031, for example to produce a calibration curve. A value of the control voltage 1031 corresponding to the target power may then be determined. In one example, the DC supply 118 may supply 4.2V and to achieve a target power of 20 W the controller 1001 may determine in an example calibration a value for the duty cycle of the PWM\_DAC signal setting of around 344 out of 800.

[0189] In one example, the controller 1001 is configured to set the control voltage 1031 at an initial value which is based on this determined value of the control voltage 1031. For example, the initial value of the duty cycle of PWM\_DAC which determines the control voltage 1031 may be set at half of the determined value corresponding to the target power. For example, where the duty cycle setting for the control voltage 1031 found to correspond with the target power is 344 out of 800, the controller 1001 may begin the session with the setting being set at 152 out of 800, and increase the setting by a predetermined amount with every predetermined interval until the measured power P is within the target range. This may have the effect that at the start of a usage session, the power delivered is well below the target power and the power delivered may then ramp up (by ramping up by the controller 1001 of the control voltage 1031) until it reaches the target power range. This initial ramping up of the power delivered may provide for improved safety in operation of the circuit 600, preventing overheating of the susceptor at the start of a session and allowing the circuit 600 to respond to the actual power supplied as determined by the controller 1001.

[0190] In one example, the predetermined interval is the same predetermined interval as is used by the controller 1001 in the method 1050 of determining which of the first 124 and second 126 inductors to activate. In one such example, as mentioned above, the predetermined intervals

are of length  $\frac{1}{64}$ s. The length of the predetermined interval (or equivalently the interrupt rate) may be chosen to provide an advantageous time interval at which the controller can monitor the circuit and adjust parameters accordingly. For example, an interrupt rate of 64 Hz, or within a range of approximately 10-100 Hz may be used. At these example interrupt rates, the controller 1001 may measure increases in temperature of the susceptor zones at a sufficiently high rate that it may determine to stop heating by a particular inductor 124, 126 before a zone 132a, 132b of the susceptor 132 can increase too far above its target temperature. Similarly, examples given for the interrupt rate may provide an advantageous frequency at which the control voltage 1031 may be adjusted to allow appropriate control of power supplied to the inductors 124, 126 to within a safe target range.

[0191] In an example method of operation of the circuit 600, a target power for use by the controller 1001 in controlling power delivered to the circuit 600 is predetermined based on characteristics of a planned usage session. For example, the target power range may be adjusted throughout a usage session.

[0192] FIG. 16 shows a schematic example of a temperature profile target1 for a portion of a usage session, which in this example is a target temperature for the single susceptor zone 132a. In this example, initially at a first part 1201 of the portion of the usage session, the first zone 132a is substantially below its target temperature target1. At this first part 1201, the circuit 600 is operating to bring the first zone 132a up to the target temperature target1. At such an example part of the usage session, a target power P1 may have a range of values of, for example, 20-21 W. The target power during the first part 1201 of the session may be relatively high in order to bring the susceptor 132, and therefore the aerosolizable material 110a, up to temperature quickly to a temperature suitable for producing aerosol for inhaling by the user.

[0193] As the usage session progresses, the first zone 132a substantially reaches its target temperature target1. A second part 1202 of the usage session may be defined beginning shortly after the first zone 132a reaches its target temperature target1. For instance, for this part 1202 of the usage session, the first zone 132a may be substantially at its target temperature target1, of e.g. 250° C., and may be being maintained at the target temperature target1 according to the method 1050. Similarly, although this is not shown in FIG. 16, the second zone 132b may be being maintained at its own target temperature target2 by the method 1050 (and the target temperature target2 of the second zone 132b may define a different temperature profile to that defined by target1).

[0194] The part 1202 in the usage session after the first zone 132a substantially reaches temperature target1 may be characterized in that the controller 1001 is operating to maintain the temperature of the first zone 132a (or of both zones 132a, 132b) rather than to bring the temperature of the first zone 132a up to its target value target1, as in the first part 1201. As such, during the part 1202 of the usage session, relatively little power may be required to be supplied to the susceptor zone 132a to maintain the target temperature target1, when compared to the power required to bring the susceptor zone 132a up to the target temperature target1. At the second part 1202 of the usage session, it may be advantageous to reduce the value of target power P1 compared to its value in part 1201. In one example, the target power level P1 may be reduced from 20-21 W in part 1201

to around 15 W during part 1202 of the usage session. Reducing the target power P1 in this way may be advantageous in some examples because by using a lower level of power energy losses in the circuit may be reduced, and thus efficiency may be increased.

[0195] For a third part 1203 of the usage session, the value of the target temperature target1 is 0, i.e. the first inductor 124 is not to be activated. At this point, the target power P1 may also be reduced to 0 if the usage session has come to an end, or if the second inductor 126 is still being activated, then the target power P1 may remain at a non-zero value while the second inductor 126 is activated. Accordingly, the target power may take into account the temperature profile of both zones 132a, 132b at any one point in the usage session. If a part of the usage session, for example, requires one of the zones to be significantly increased in temperature, then a relatively high target power may be appropriate. Conversely, for parts of a usage session where neither zone 132a, 132b requires substantial heating, a relatively low target power may be used.

[0196] As mentioned above, use of lower power levels during certain periods of a usage session may provide advantages in that an energy saving may be achieved over the duration of a session. For example, where the target power level is reduced from 20-21 W in the first period to around 15 W in the second period, in some examples an energy saving of around 5-10% may be achieved due to reduced energy losses in the circuit 600 when operating at lower power. In one example, over the course of a typical session of around 260 s in length, maintaining the target power at around 20 W for the full duration of the session may result in energy usage of around 1000 J. However, reducing the target power to around 15 W upon the first zone 132a first reaching its set temperature and maintaining the target power level at 15 W for the remainder of a session of substantially the same length may result in an energy usage of between 900 and 950 J. In examples, almost all of the power used by the device is due to energy supplied to heat the susceptor 132. The power usage of electrical components other than the heating circuitry, e.g. LED indicators and the microcontroller, may be less than around 0.1 W and in some examples may be less than around 0.01 W.

[0197] The example device 100 comprises a temperature sensing arrangement for sensing a temperature of the susceptor 132. For example, the temperature sensing arrangement may comprise one or more temperature sensors, and in one example comprises one temperature sensor for each zone of the susceptor 132. In one example, as described above, the susceptor comprises a first zone 132a and a second zone 132b and the controller 1001 operates the induction heating circuit 600 of the device 100 as described above with reference to earlier figures to heat each zone by a respective inductor 124, 126.

[0198] In examples, the controller 1001 is configured to determine whether one or more criteria that are indicative of a fault with the temperature sensing arrangement are satisfied. If the controller determines that said one or more predetermined criteria are satisfied then the controller 1001 takes a control action, for example stopping or reducing the supply of energy to heat the susceptor 132 or issuing a warning signal indicating that a fault has occurred. Examples of said predetermined criteria are described below. A safety feature may therefore be provided which

allows the controller 1001 to take a control action if a fault with the temperature sensing arrangement is identified.

[0199] FIG. 17 shows further details of the susceptor 132. The susceptor 132, as described above, comprises a first zone 132a, and a second zone 132b. A first thermocouple 133a is arranged to measure the temperature of the first susceptor zone 132a and a second thermocouple 133b is arranged to measure the temperature of the second susceptor zone 132b. As will be well understood, a thermocouple is a device used for sensing temperature which comprises two dissimilar electrical conductors. The two conductors are held at the same electric potential at a first measurement end while a second end of the conductors is held at a known temperature to form a second reference end. In some examples, the ends of the two wires may be connected at the measurement end or in other examples the two wires may be connected to a single conductive surface, such as that of the susceptor 132. A voltage is generated between the conductors which is dependent on a temperature difference between the measurement end and the reference end, according to the Seebeck effect. If the temperature of the reference end is known for example through measurement by a temperature sensor, and in one example herein by use of 10 k $\Omega$  thermistor, then a temperature at the measurement end can be determined from the voltage generated between the conductors.

[0200] In the example of FIG. 17, the first thermocouple 133a comprises a first pair of wires 1704, 1705 and the second thermocouple 133b comprises a second pair of wires 1708, 1709. In one example, both thermocouples are J-type thermocouples, that is, a first wire 1704, 1708 of each pair is formed of constantan while a second wire 1705, 1709 of each pair is formed of iron. In other examples, different types of thermocouples comprising different pairs of dissimilar conductors may be used, such as type E, K, M thermocouples for example, or a different type of thermocouple may be used for each susceptor zone 132a, 132b.

[0201] The first thermocouple 133a comprises a first measurement junction 1706 at which the constantan wire 1704 and iron wire 1705 of the first thermocouple 133a are joined together and which is attached to the first susceptor zone 132a. In this example, the first measurement junction 1706 is attached to the first susceptor zone 132a by spot-welding to the surface of the susceptor 132. Similarly, the second thermocouple 133b comprises a second measurement junction 1310 at which the constantan wire 1708 and iron wire 1709 of the first thermocouple 133a are joined together and which is spot-welded to the susceptor 132 at the second susceptor zone 132b. Insulating sheaths 1707, 1711 cover each wire of the first thermocouple 133a and second thermocouple 133b. The first thermocouple 133a terminates at a pair of terminals 1704a, 1705b for providing reference end voltages for the wires 1704, 1705 respectively to the controller 1001, allowing determination of the temperature at the first measurement junction 1706 by the controller 1001. Similarly, the second thermocouple 133b terminates at a pair of terminals 1708a, 1709a for providing reference end voltages to the controller 1001 for the wires 1708, 1709 respectively and allowing determination of the temperature at the second measurement junction 1710 by the controller 1001. Each of the terminals 1704a, 1705a, 1708a, 1708a may be connected to respective input pins of the controller 1001, or may be attached to the PCB 122 and thereby to the controller 1001. For example the terminals 1704a, 1705a,

1708a, 1708a may each be soldered to the PCB 122, to provide an input to the controller 1001.

[0202] Another example of a temperature sensing arrangement for the susceptor 132 is shown in FIG. 18. In this example the temperature sensing arrangement again comprises two thermocouples each comprising a constantan wire. However, in this example the two thermocouples share an iron wire. That is, a first thermocouple 183a is for allowing the controller 1001 to determine a temperature of the first susceptor zone 132a and comprises a first constantan wire 1804 attached to the first susceptor zone 132a at a measurement end 1806. An iron wire 1805 forms a second wire of the first thermocouple 183a and is attached at the second susceptor zone 132b in this example. A second thermocouple 183b for measuring the temperature of the second susceptor zone 132b comprises a first constantan wire 1808 attached to the second susceptor zone 132b at a point 1810. The iron wire 1805 also forms the second wire of the second thermocouple 132b. The first constantan wire 1804 of the first thermocouple 132a comprises a reference end 1804a, the first constantan wire 1808 of the second thermocouple 132b comprises a reference end 1808a, and the iron wire 1805 forming the second wire of both the first and second thermocouples 183a, 183b comprises a reference end 1805a. The reference ends 1804a, 1805a, 1808a provide respective reference end voltages for each of the wires 1804, 1805, 1808 to the controller 1001 to allow the controller 1001 to determine temperatures of the first and second susceptor zones 132a, 132b. That is, the reference voltages provided by the constantan wire reference end 1804a of the first thermocouple 183a and the reference voltage provided by the iron wire reference end 1805a allow the controller 1001 to determine the temperature of the first susceptor zone 132a. Similarly, the reference voltages provided by the constantan wire reference end 1808a of the second thermocouple 183b and the reference voltage provided by the iron wire reference end 1805a allow the controller 1001 to determine the temperature of the second susceptor zone 132b. As described with reference to the example shown in FIG. 17, the junction ends 1804a, 1805a, 1808a of the wires may be attached to the PCB 122 to provide reference voltages to the controller 1001. In the arrangement shown in FIG. 18, the susceptor 132 is iron. This allows for the use of the common iron reference wire 1805, since, because the iron wire 1805 and the susceptor 132 are made of the same material, the point at which the iron wire 1805 joins to the susceptor 132 is not a thermocouple junction. Providing three wires, as in the arrangement shown in FIG. 18, may be advantageous when compared to providing four wires, as in the arrangement of FIG. 17, in that it requires fewer wires to be attached, e.g. soldered, to the PCB 122.

[0203] FIG. 19 shows a schematic representation of apparatus 1900 for providing control functions relating to temperatures in the device 100. As can be seen from FIG. 19, the apparatus 1900 is arranged to receive the reference voltages provided by the reference ends 1804a, 1805a, 1808a of the thermocouple wires 1804, 1805, 1808 shown in the temperature sensing arrangement example of FIG. 18. Describing first the input from the constantan wire 1804 of the first thermocouple 183a, the constant wire reference end 1804a of the first thermocouple 183a is attached to the apparatus 1300 at a point P8. From there, the reference voltage signal at 1804a is arranged to be supplied to the positive and negative input terminals of a first thermocouple op-amp



U4A (component type OPA2376). The point P8 is connected to the positive terminal of the op-amp U4A via a 2.49 k $\Omega$  resistor R26 and a 2.2 nF capacitor C20. The point P8 is connected to the negative terminal of the op-amp U4A via the resistor R26 and a second 2.49 k $\Omega$  resistor R27. The op-amp U4A is connected in a closed-loop configuration with 536 k $\Omega$  resistor R23 and a 2.2 nF capacitor C19 arranged in parallel between the output and the negative terminal of the op-amp U4A. The positive terminal of the op-amp U4A is also connected to ground. The op-amp U4A is powered by a 3.8V supply, which is provided by the controller 1001 and which is connected to a power input of the op-amp via a 100 $\Omega$  resistor R17. The power input of the op-amp U4A is connected to ground via a 10 nF capacitor C14.

[0204] The first thermocouple op-amp U4A is configured to receive a signal from the first thermocouple constantan wire 1804 at the point P8 and provide amplify this signal for outputting to the controller 1001. In this example, the op-amp U4A provides a gain of 107.63 to the input signal from thermocouple reference end 1804a. This gain may be provided by selecting appropriate values for the resistors R26, R27 and R23 for example. In selecting a gain for the op-amp U4A, the range of temperatures to be measured by the thermocouple 183a, the reference voltage produced by a J-type at that range of temperatures and a suitable voltage range to be provided to the controller 1001 may be taken into account. In one example, a thermocouple reference voltage of 23.228 mV corresponds to a temperature reading of 425° C. To amplify this reference voltage to a value of 2.5V to provide to the controller 1001 requires a gain of 107.63, as provided by the example component values shown in FIG. 19. An amplified first thermocouple voltage TC1 is provided to the controller 1001 at a point 1901 which is connected to the output of the op-amp U4A via a 1 k $\Omega$  resistor R22. The point 1901 is connected to ground via a 100 nF capacitor C21. From the voltage signal TC1 provided at the point 1901 the controller 1001 is configured to determine a temperature value measured by the first thermocouple 183a, corresponding to the temperature of first susceptor zone 132a.

[0205] The second thermocouple 183b is, as described above, also a J-type thermocouple and provides a second thermocouple reference voltage TC2 to the controller 1001 at a point 1902 in the same manner as the first thermocouple reference voltage TC1 is provided to the controller 1001. This allows the controller 1001 to determine a temperature for the second susceptor zone 132b in the same manner as described for the first thermocouple 183b. It will be understood that a plurality of resistors R37, R38, R39, R40 and capacitors C30, C31, C32 connected to the second op-amp U4B provide the equivalent function as the resistors R22, R23, R26, R27 and capacitors C19, C20, C21 described above in connection with the first thermocouple op-amp U4A.

[0206] In addition to providing to the controller 1001 the amplified thermocouple voltages TC1, TC2, the apparatus 1900 is also configured to provide a safety feature in the event that a temperature measured by either of the thermocouples 183a, 183b exceeds a given threshold. That is, the output of the first thermocouple op-amp U4A is connected to a negative terminal input of a first precision comparator U3A (component type AS393). A positive input of the precision comparator U3A is connected to a 2.5V signal via a 3.3 k $\Omega$  resistor R42 and to ground via a 10 k $\Omega$  resistor

R43. The precision comparator U3A is configured such that if the amplified thermocouple voltage received from the op-amp U4A indicates a temperature exceeding a particular value, the first precision comparator U3A outputs a stop signal /FF RST to stop heating of the susceptor 132 by the induction heating circuit 600. In examples, the signal /FF RST is sent to the flip-flop 622 input RST, to reset the output value Q of the flip-flop 622, in the manner described above with reference to earlier figures, to thereby stop heating by the circuit 600. In one example the threshold temperature value at which the apparatus 1900 is configured to send the signal to stop heating by the circuit 600 is 280° C.

[0207] A second precision comparator U3B performs an equivalent function for the second thermocouple 183b, receiving an input at its negative terminal from the second thermocouple op-amp U4B and being configured to send the signal /FF RST if the received signal from the op-amp U4B indicates that the second thermocouple 183b is measuring a temperature exceeding the threshold value.

[0208] A third comparator U6 is also included in FIG. 19 and provides for another mechanism by which the signal /FF RST may be sent to stop heating of the susceptor 132. The comparator U6 receives a signal COIL TEMP from a thermistor (not shown) located on the insulating member 128 which indicates a temperature at the outside of the insulating member 128, proximate the coils 124, 126. The comparator U6 is configured to compare this signal COIL TEMP to a threshold value and provide a signal, in this example a signal /FF RST to the flip-flop 622 to stop operation of the circuit 600 to heat the susceptor 132 if the comparator U6 determines that the value of the temperature represented by COIL TEMP is too high.

[0209] It will be appreciated that for accurate measurement of the temperatures of the susceptor zones 132a, 132b, each measurement ends (1706, 1710 in the example of FIG. 17, or 1806, 1810, 1805 in the example of FIG. 18) should be at a temperature which is substantially the same as the respective temperature of the zone 132a, 132b for which the temperature is being measured. As such, good thermal conductivity between the susceptor 132 and the thermocouples 183a, 183b should be maintained.

[0210] To describe this with reference to an example, it will be appreciated that if, for example, the attachment point 1806 of the first constantan wire 1804 were to become detached from the first susceptor zone 132a then thermal conduction between that susceptor zone 132a and the first thermocouple 183a would be lost or at least significantly reduced. In such a circumstance, the first thermocouple 183a may, for example, remain detached from but within close proximity of the susceptor 132 (and may continue to be heated by the susceptor 132 to some extent by e.g. radiative or convective processes). However, since heat would no longer be properly conducted between the susceptor 132 and the detached first thermocouple 183a, the temperature of the first susceptor zone 132a (which the first thermocouple 183a is intended to measure) and the detached thermocouple wire 1804 may diverge as the first susceptor zone 132a is heated. In this circumstance, the first thermocouple 183a would not, for example, allow the controller 1001 to accurately determine increasing temperature of its corresponding first susceptor zone 132a as said zone 132a is heated. If heating of the first susceptor zone 132a were to continue in this manner without its temperature being properly measured by use of the first thermocouple 183a, a danger of overheating of the

susceptor **132** may arise. It will be appreciated that the same principle described above also applies to the second susceptor zone **132b** in the circumstance that the second thermocouple **183b** were to become decoupled therefrom.

[0211] In one example, the device **100** is configured to cut off heating of the susceptor **132** in the event of an indication that either of the susceptor thermocouples **183a**, **183b** is not correctly measuring the temperature of the susceptor **132**. This may be advantageous in that a safety feature is provided which allows cutting off of power to heat the susceptor **132** when the controller **1001** determines that the temperature of the susceptor **132** is not being properly measured.

[0212] In one example, as described above in relation to the induction heating circuit **600** of the device **100**, a power supplied to the induction heating circuit for heating of the susceptor **132** may be determined by measuring the DC voltage supplied by the DC supply and the DC current supplied by the DC supply and calculating a product of these values to obtain a power. From this measurement of power, the controller **1001** may be configured to determine an amount of energy supplied to the circuit over a given period of time. As above, the controller **1001** is also configured to receive temperature measurements from the thermocouples **183a**, **183b**. The controller **1001** is configured to determine a change in temperature measured by the thermocouples **183a**, **183b** over the given period of time. In one example then, the controller **1001** may determine the amount of energy supplied per  $1^{\circ}\text{C}$ . rise in temperature measured by one of the susceptor thermocouples **183a**, **183b**. For example, the controller **1001** may determine a ratio of the amount of energy supplied to the first susceptor zone **132a** over a given time period to the increase in the temperature measured by the first thermocouple **183a** over that time period. Similarly, the controller **1001** may determine the amount of energy supplied to the second susceptor zone **132b** per  $1^{\circ}\text{C}$ . rise in temperature measured by the second thermocouple **183b**—in examples this amount of energy per  $^{\circ}\text{C}$ . rise is measured temperature may differ to that for the first zone due to differing properties of the first and second inductors **124**, **126**.

[0213] In one example, during setup of the device **100** for example, the amount of energy supplied to a given one of the inductors **124**, **126** for heating the corresponding susceptor zone **132a**, **132b** of that inductor which results in a measured temperature rise by the corresponding thermocouple **183a**, **183b** of a given amount may be determined. That is, through testing or otherwise, it may be determined that in normal operation when, for example, the first thermocouple **183a** is properly attached to the first susceptor zone **132a**, on average, supplying 400-1000 mJ, or around 500 mJ to the first inductor **124** results in a temperature rise measured by the first thermocouple **183a** of  $1^{\circ}\text{C}$ . That is, the ratio of energy supplied to measured increase in temperature may be 400-1000 mJ/ $^{\circ}\text{C}$ ., or around 500 mJ/ $^{\circ}\text{C}$ . In another example, it may be determined that at any point during normal operation of the device, the maximum value of the ratio of energy supplied to the first inductor **124** to rise in temperature measured by the first thermocouple **183a** is 500 mJ/ $^{\circ}\text{C}$ .

[0214] However, if, as described above, the first thermocouple **183a** becomes thermally decoupled from the first susceptor zone **132a**, then the amount of energy supplied to the first inductor **124** which results in a  $1^{\circ}\text{C}$ . temperature rise being measured by the first thermocouple **183a** may significantly increase. For example, through testing it may

be found that if the first thermocouple **183a** is decoupled from the first susceptor zone **132a** then the ratio of the energy, as measured in mJ, supplied to the first inductor **124** to the increase in temperature measured by the first thermocouple **183a** may be around a very high number, such as 40,000 to 100,000 J/ $^{\circ}\text{C}$ . That is, if the first thermocouple **183a** becomes detached from the susceptor zone **132a** and is no longer properly measuring its temperature, then the thermocouple may record only small increases in temperature, despite the susceptor **132a** being heated by the inductor **124**. The controller **1001** may thus determine that if the ratio of energy supplied to heat the first susceptor zone **132a** to the temperature rise measured by the first thermocouple **183a** is greater than a given amount, then the first thermocouple **183a** may no longer be correctly measuring the temperature of the first susceptor zone **132a**, e.g. the first thermocouple **183a** may have become detached from the susceptor **132**. Similarly, the controller **1001** may determine that if a ratio of energy supplied to heat the second susceptor zone **132b** to the temperature rise measured by the second thermocouple **183b** is greater than a given amount, then the second thermocouple **183b** may no longer be correctly measuring the temperature of the second susceptor zone **132b**.

[0215] In this example, the controller **1001** is configured to stop the supply of power to heat the susceptor **132** if the ratio of energy supplied to heat the susceptor **132** to the temperature rise measured by one of the first and second thermocouples **183a**, **183b** is greater than a predetermined amount. Cutting-off of the power for heating the susceptor **132** in this manner may act as a safety feature which helps prevent over-heating of the susceptor **132** where one (or both) of the thermocouples **183a**, **183b** has become detached from the susceptor **132** or is otherwise no longer providing accurate susceptor temperature measurements. The controller **1001** may, for example, stop the supply of energy to heat the susceptor **132** by, deactivating the circuit **600** by use of a signal sent to the control arrangement of the induction heating circuit **600**. In another example, the controller **1001** may take another control action in response to determining that the ratio is greater than the predetermined amount. For example, the controller **1001** may reduce the power supplied to the circuit **600** to heat the susceptor **132**. For example, the controller **1001** may reduce the control voltage **1031** to reduce power supplied to heat the susceptor **132**.

[0216] In one example, an expected measured value of the ratio of energy to heat one of the susceptor zones **132a**, **132b** to the temperature rise measured by the corresponding thermocouple **183a**, **183b** in normal operation may be around 500 mJ/ $^{\circ}\text{C}$ . and the controller **1001** may be configured to stop the supply of power to heat the susceptor **132** if it determines that the ratio has become greater than or equal to a example value from 2000 mJ/ $^{\circ}\text{C}$ . to 4000 mJ/ $^{\circ}\text{C}$ . As such, in this example, a margin, in this case of around 1500 mJ/ $^{\circ}\text{C}$ . to 3500 mJ/ $^{\circ}\text{C}$ . above the expected value is provided which may allow for minor increases of the ratio above the expected level to occur without the controller **1001** taking a control action such as cutting off the power supply.

[0217] In examples, during operation of the device **100** a value for the ratio between the energy supplied to heat the susceptor **132** to the measured increase in temperature sensed by one of the thermocouples **183a**, **183b** may be determined at predetermined intervals. The predetermined

intervals may be, for example, the same as those used for controlling the induction heating circuit 600, such as those described above, and may in one example be of length  $\frac{1}{64}s$ . Where one of the thermocouples 183a, 183b has become detached from the susceptor 132, it has been found that in one example a ratio of  $4000 \text{ mJ}/^\circ\text{C}$ . for that thermocouple is exceeded within around 0.5 s. As such, in this example where the cut-off ratio is set at  $4000 \text{ mJ}/^\circ\text{C}$ ., the supply of power may be cut-off within around 0.5 s, and thus substantial overheating of the susceptor 132 may be avoided.

[0218] FIG. 20 shows a flowchart representation of an example method 1400 of controlling the device 100. In an example the method 1400 is performed by the controller 1001. At block 1402 power is supplied from the power supply to heat the susceptor 132. At block 1404 an energy  $\Delta E$  supplied to the heating circuit to heat the susceptor 132 over a given time period is determined. As described above, at any one time, power is only supplied to one or the other of the first inductor 124 and the second inductor 126 during the given time period. As such, the power measured at block 1404 is a power supplied to the one of the first inductor 124 and second inductor 126 which is active during the given time period. This may be determined as described above, for example, by measuring a DC voltage and a DC current supplied to the heating circuit by the DC supply and calculating a product of these values. For example, an energy used over the time period may be determined by determining an average power over the time period and multiplying this by the duration of the time period, e.g. of  $\frac{1}{64}s$ .

[0219] At block 1406 a change in temperature  $\Delta T$  measured by one of the susceptor thermocouples 183a, 183b over the time period is determined. The temperature change  $\Delta T$  will in examples correspond to the temperature change measured by the thermocouple corresponding to the inductor to which power was supplied during the given period. That is, for example, if during the given time period power is supplied to the first inductor 124, then the energy  $\Delta E$  is the energy supplied to the first inductor 124 during the given time period and the temperature change  $\Delta T$  is the temperature change recorded by the first thermocouple 183a during the given time period. At block 1408 a ratio  $\Delta E/\Delta T$  of the energy  $\Delta E$  supplied over the time period to the change in the temperature  $\Delta T$  measured by the respective susceptor thermocouple 183a, 183b over the given time period is determined. At block 1410 the ratio  $\Delta E/\Delta T$  is compared to a cut-off value. The cut-off value is a threshold value for  $\Delta E/\Delta T$  which may be predetermined during setup of the device 100 as discussed above. The cut-off value may be greater than an expected value for the ratio  $\Delta E/\Delta T$  when the device 100 is operating normally. The cut-off value may be greater than the expected value by a given margin, e.g. a margin of around  $500 \text{ mJ}/^\circ\text{C}$ ., or around  $1000 \text{ mJ}/^\circ\text{C}$ .

[0220] If, at block 1410 the controller 1001 determines that the determined value for the ratio  $\Delta E/\Delta T$  is less than the cut-off value then the method returns to block 1402 and the controller 1001 continues to allow power to be supplied to heat the susceptor 132 for another instance of the given time period. If, however, at block 1410 the controller 1001 determines that the ratio  $\Delta E/\Delta T$  for the given time period is greater than or equal to the cut-off value then the method proceeds to block 1412.

[0221] At block 1412 the controller 1001 determines if the inductor 124, 126 has been active to heat the susceptor 132 for a predetermined length of time, of for example 0.2-1 s or

around 0.5 s. The predetermined length of time may, for example comprise a number of consecutive predetermined number of time periods. For example, at block 1412 the controller 1001 may determine if the circuit 600 has been active to heat the susceptor 132 for 32 consecutive time periods of  $\frac{1}{64}s$ , totaling a length of time of 0.5 s. If the answer is “no” then despite the determined ratio  $\Delta E/\Delta T$  at block 1410 being above the cut-off value the method returns to block 1402 and the controller 1001 continues to allow heating of the susceptor 132. The step performed at block 1412 allows the method 1400 to take into account that when the controller 1001 is operating to maintain the temperature of e.g. the first susceptor zone 132a, rather than acting to increase the temperature of said zone 132a, energy may be supplied to heat the zone 132a while relatively small increases in temperature are recorded by the first thermocouple 183a. Therefore, a value for  $\Delta E/\Delta T$  above the cut-off value may be obtained which does not necessarily indicate a fault with the operation of the thermocouple 183a to measure the temperature of the first susceptor zone 132a.

[0222] In another example, the predetermined period over which  $\Delta T$  and  $\Delta E$  are calculated may be the same as the length of time used at block 1412. That is, in one example,  $\Delta T$  is the change of measured temperature over a 0.5 s period and  $\Delta E$  the energy supplied over that 0.5 s period. In this example, therefore, at block 1412 the controller 1001 determines if the circuit 600 has been active to heat the susceptor 132 for the entirety of the previous 0.5 s time interval and if the answer is “yes” the controller 1001 proceeds to block 1414. In this example, it will be understood that during the 0.5 s period the controller 1001 may determine once every  $\frac{1}{64}s$  which of the inductors 124, 126 if either should be active to heat the susceptor 132 and thus block 1412 may amount to the controller 1001 determining that one of the inductors 124, 126 was active for each of the 32 previous  $\frac{1}{64}s$  intervals.

[0223] At block 1414, the controller 1001 cuts off the supply of power to heat the susceptor 132. This acts as a safety mechanism which may prevent over-heating of the susceptor 132 in the event that one or both of the thermocouples 183a, 183b is not properly measuring changes in temperature of the susceptor 132 for a given length of time.

[0224] As described in examples above, the controller 1001 is configured to implement the described safety feature by determining if a predetermined criteria is satisfied based on a temperature of the susceptor 132 and the energy supplied to the circuit 600. However, it will be appreciated that in other examples, the controller 1001 may determine whether to take a control action such as stopping the supply of energy to the heating circuit based on one or more predetermined criteria using a different temperature measured in the device 100. For example, the above-described method of determining, based on a comparison between a measured increase in the temperature and an amount of energy supplied to the heating circuit, if a criteria is met which indicates a fault in measuring a temperature in the device 100 may be used elsewhere in the device 100.

[0225] As described above, the controller 1001 of the example induction heating device 100 is configured to perform a method of stopping supply of energy to heat a susceptor if one or more predetermined criteria that are indicative of a fault with measurement of a temperature of the susceptor are satisfied. However, in other examples, the above-described method may be applied by a controller in a

different type of aerosol generating device than a device which comprises an inductive heating circuit. For example, the above-described example method may be applied in an aerosol generating device comprising a heating circuit which is not an inductive heating circuit. In such an example the heating circuit may comprise one or more resistive heating elements for generating heat to heat an aerosol generating material when a current is passed through said resistive heating elements. In an example a temperature sensing arrangement may comprise temperature sensors for sensing temperatures of one or more such a resistive heating elements and such temperature sensors may be attached to or in close proximity to said heating elements during normal use. The controller of such a device may be configured to determine a fault with the arrangement, such as detaching of one of the temperature sensors from the heating element, by methods described above with reference to FIG. 20.

[0226] Examples above have described the controller **1001** determining a predetermined criteria which is dependent on a measured increase in temperature of the thermocouples **183a**, **183b**. In other examples, the controller **1001** may, for example, determine that a fault has occurred with the temperature sensing of the susceptor **132** by the thermocouples **183a**, **183b** if the controller determines that energy has been supplied to heat the susceptor **132** over a given period while a reduction in temperature has been measured by the thermocouples **183a**, **183b** over that same period.

[0227] Certain methods described herein may be implemented by way of non-transitory computer program code that is storable on a non-transitory storage medium. For example, in certain examples, the controller **1001** may comprise a non-transitory computer readable storage medium comprising a set of computer-readable instructions stored thereon and a processor to perform a method described herein when executed by the controller **1001**. The controller **1001** may comprise one or more processors. For example, in some examples, as described above, the controller **1001** is a programmable micro-processing unit. The controller **1001** may comprise a storage medium comprising a set of machine readable instructions, e.g. in the form of computer code, which when executed by the controller **1001** cause a method described herein to be performed.

[0228] While some of the example circuits described herein make use of silicon FETs for certain switching functions other suitable components may be used in place of such FETs. For example, components comprising wide bandgap materials such as silicon carbide, SiC, or gallium nitride, GaN, may be used. Such components may in some examples be FETs but in other examples may be high electron mobility transistors (HEMT). Such components may be faster and have higher breakdown voltages than silicon FETs which may be advantageous in some examples.

[0229] The above embodiments are to be understood as illustrative examples of the invention. Further embodiments of the invention are envisaged. It is to be understood that any feature described in relation to any one embodiment may be used alone, or in combination with other features described, and may also be used in combination with one or more features of any other of the embodiments, or any combination of any other of the embodiments. Furthermore, equivalents and modifications not described above may also be employed without departing from the scope of the invention, which is defined in the accompanying claims.

What is claimed:

1. Apparatus for an aerosol generating device, the apparatus comprising:

- a heating circuit for heating a heating arrangement, the heating arrangement arranged to heat an aerosol generating material to thereby generate an aerosol;
- a temperature sensing arrangement for sensing a temperature of the device; and
- a controller for controlling a supply of energy to the heating circuit, wherein the controller is configured to:
  - determine a characteristic that is indicative that energy is being supplied to the heating circuit over a given time period, and determine a change in temperature sensed by the temperature sensing arrangement over the given time period; and
  - take a control action if, based on the determined characteristic and the increase in temperature sensed by the temperature sensing arrangement over the given period, the controller determines that one or more predetermined criteria that are indicative of a fault with the temperature sensing arrangement are satisfied.

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